

BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



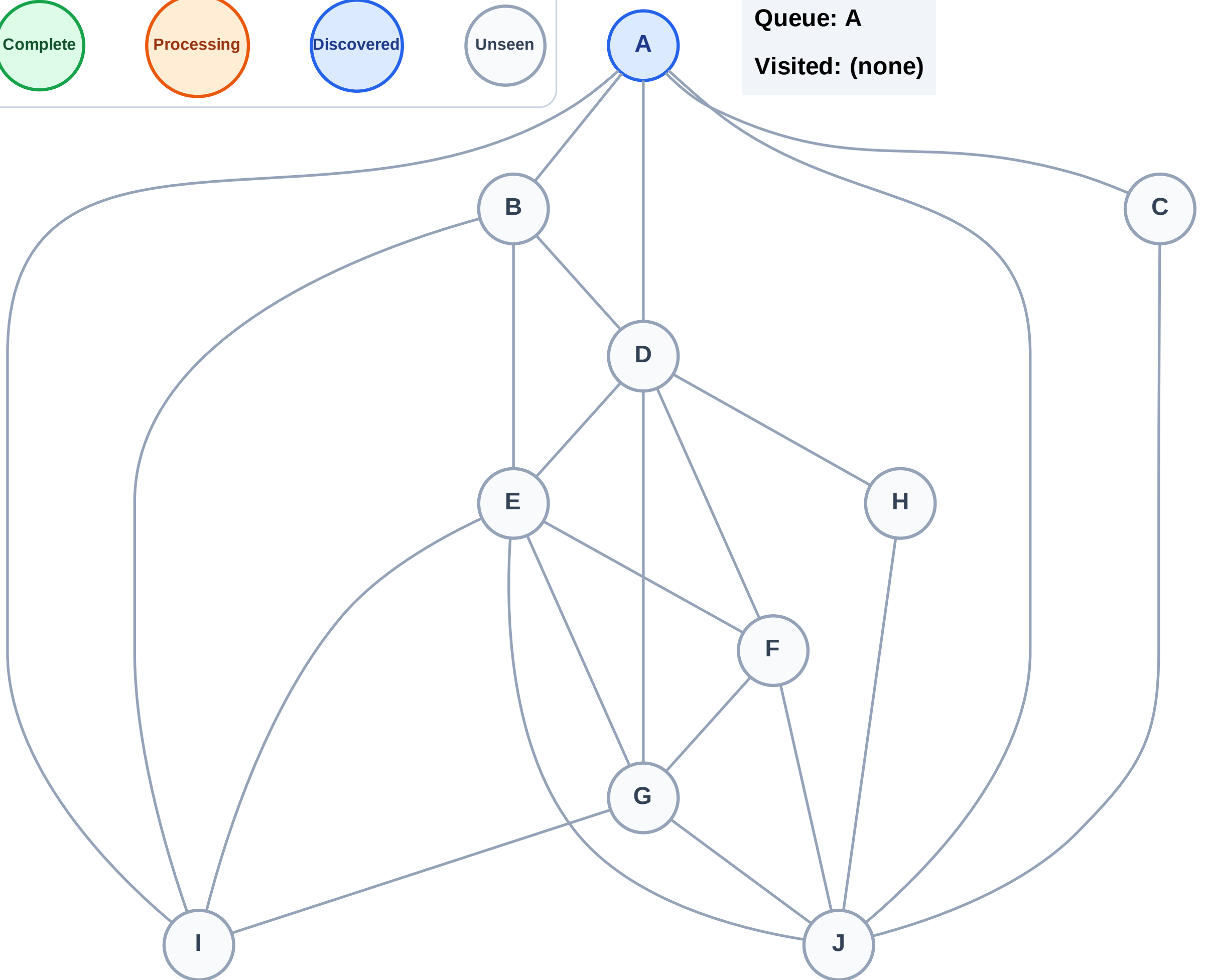
Discovered



Unseen

Queue: A

Visited: (none)



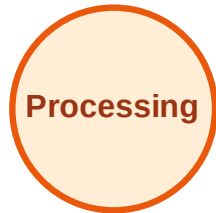
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



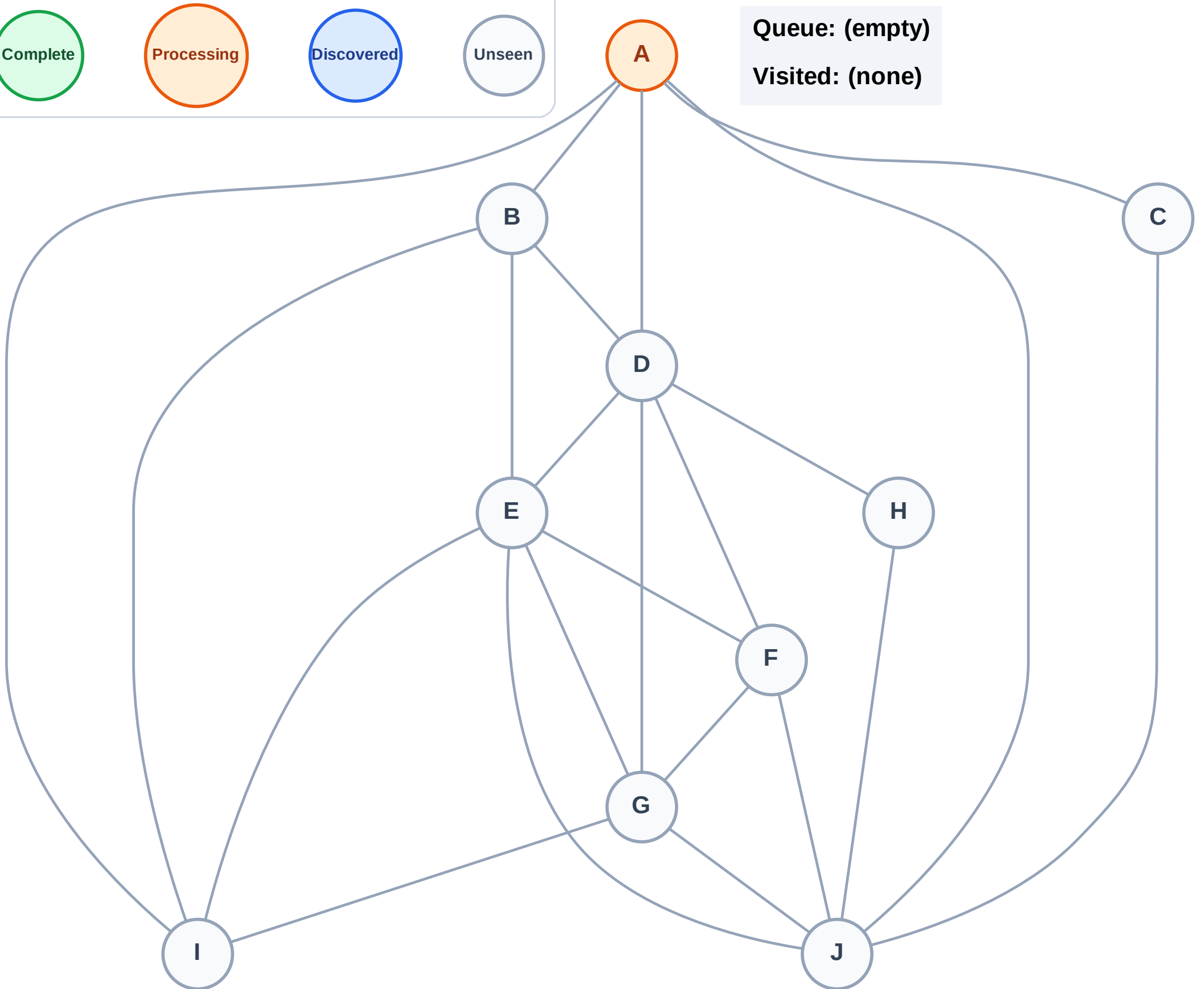
Discovered



Unseen

Queue: (empty)

Visited: (none)



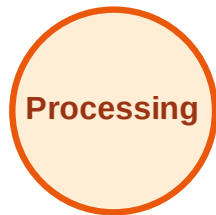
BFS Traversal

Step 3: Discover B, C, D, I, J from A

Legend



Complete



Processing



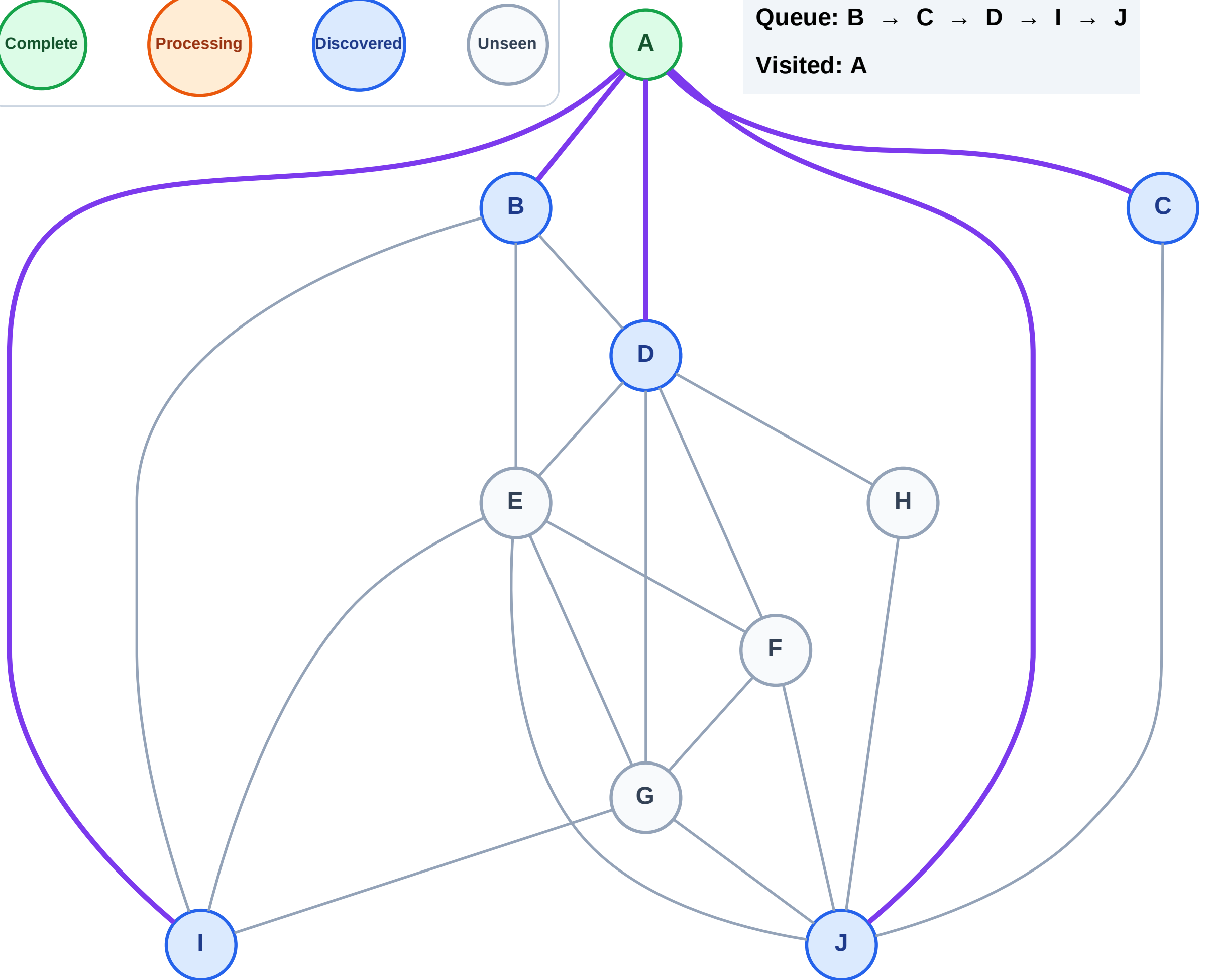
Discovered



Unseen

Queue: B → C → D → I → J

Visited: A



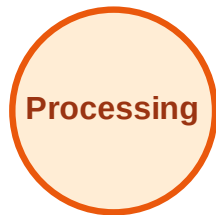
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



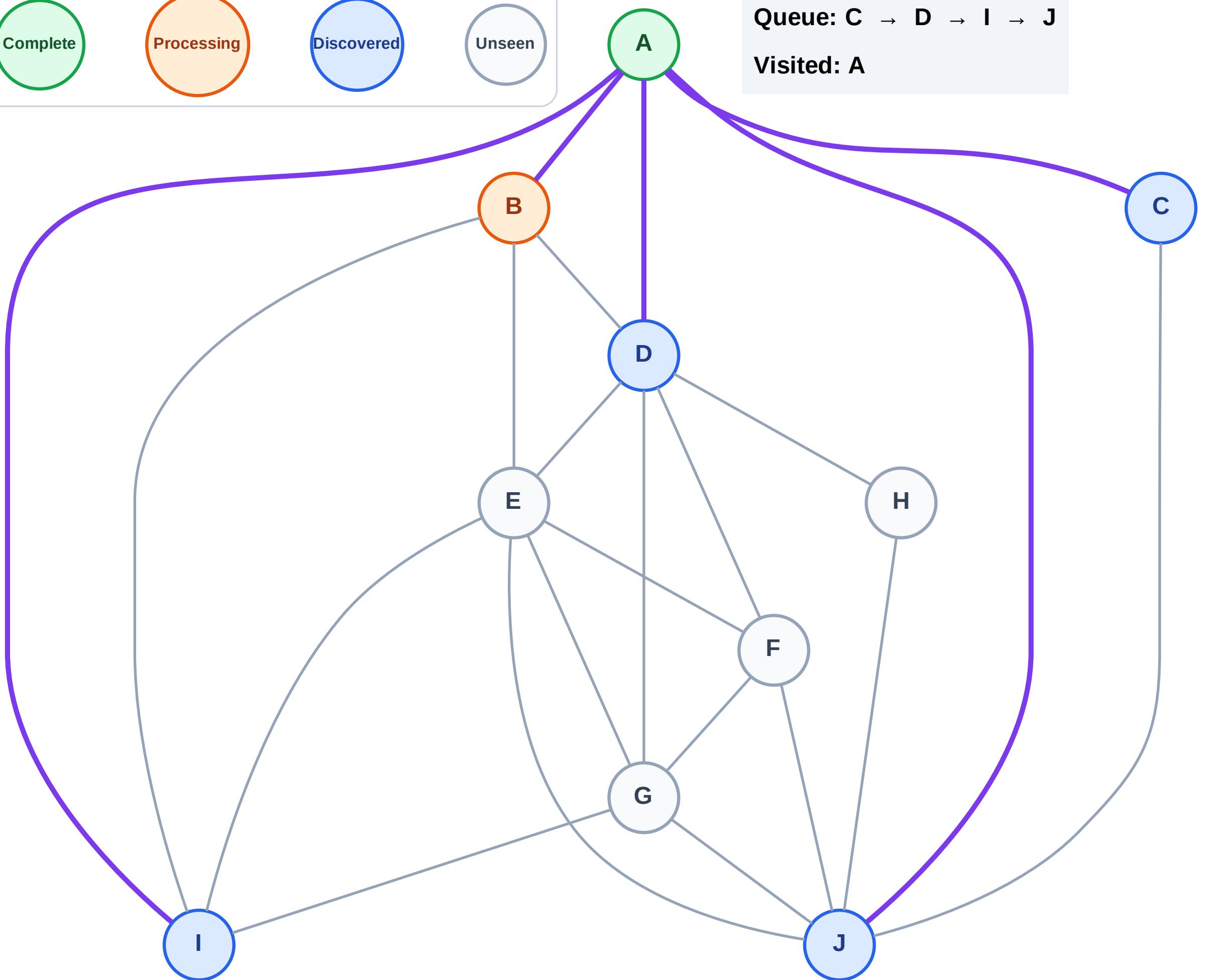
Discovered



Unseen

Queue: C → D → I → J

Visited: A



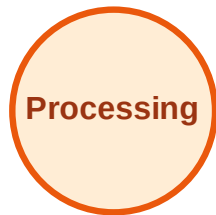
BFS Traversal

Step 5: Discover E from B

Legend



Complete



Processing



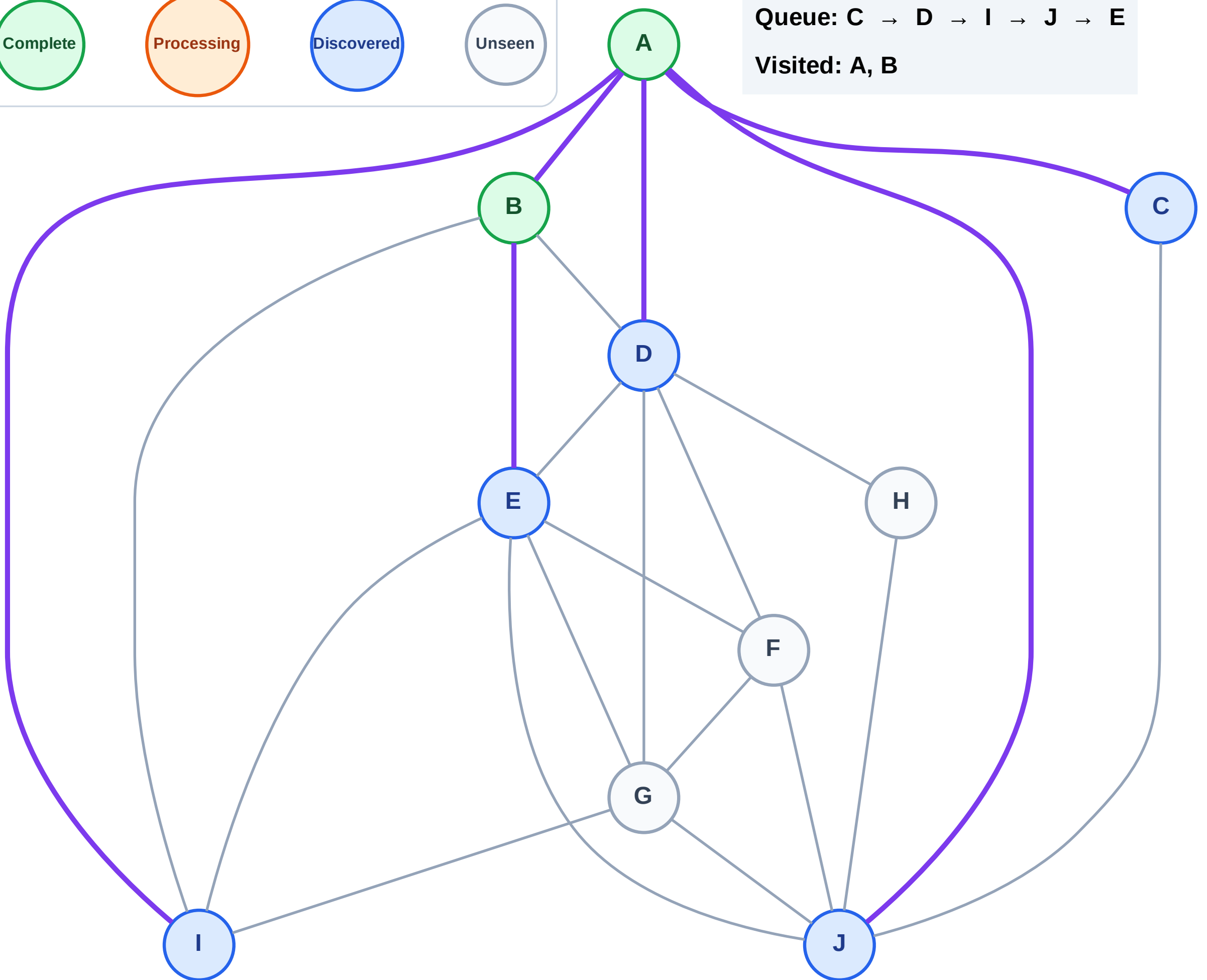
Discovered



Unseen

Queue: C → D → I → J → E

Visited: A, B



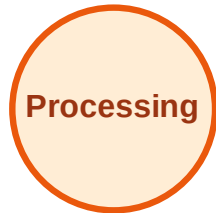
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



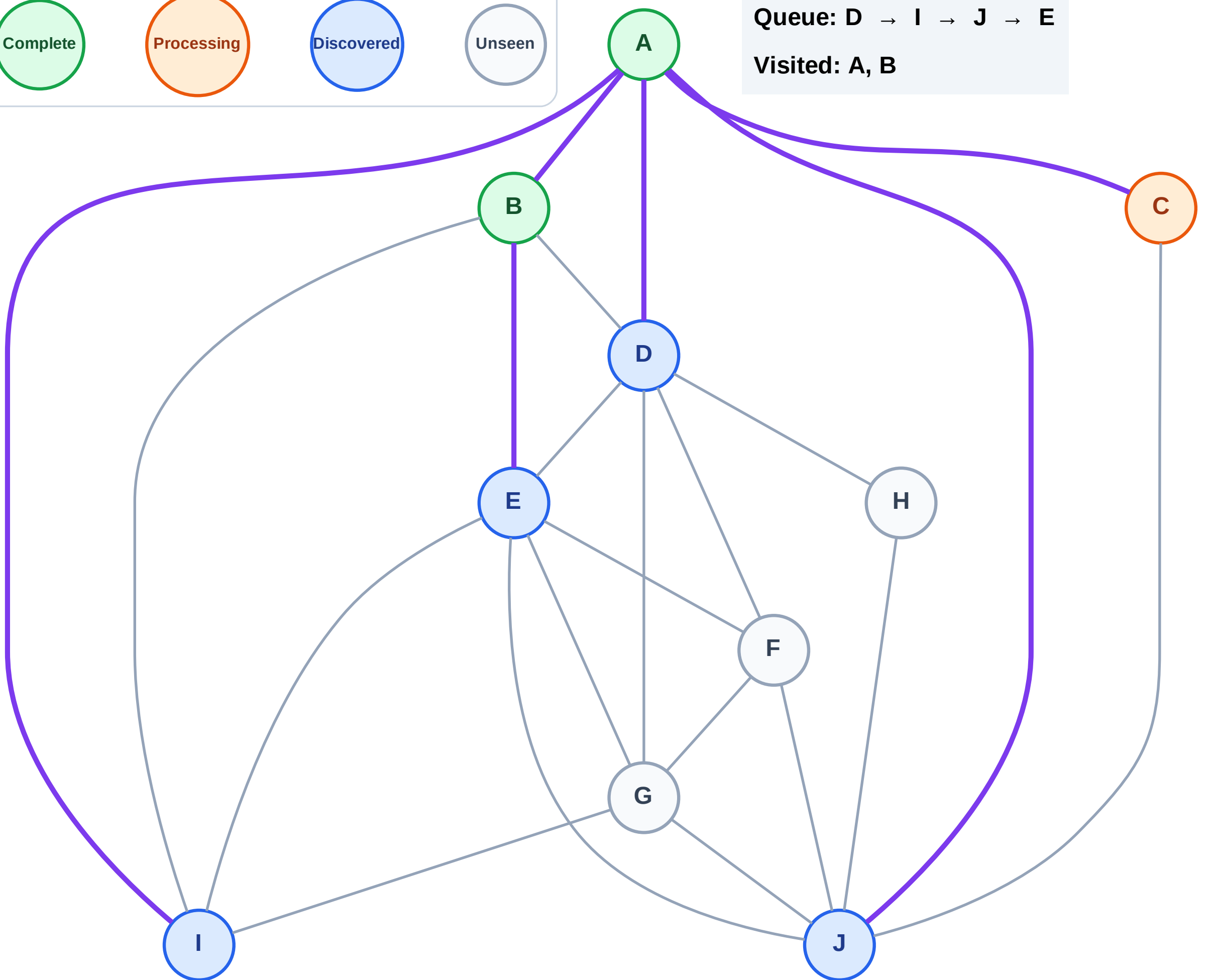
Discovered



Unseen

Queue: D → I → J → E

Visited: A, B



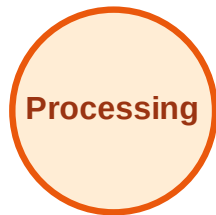
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



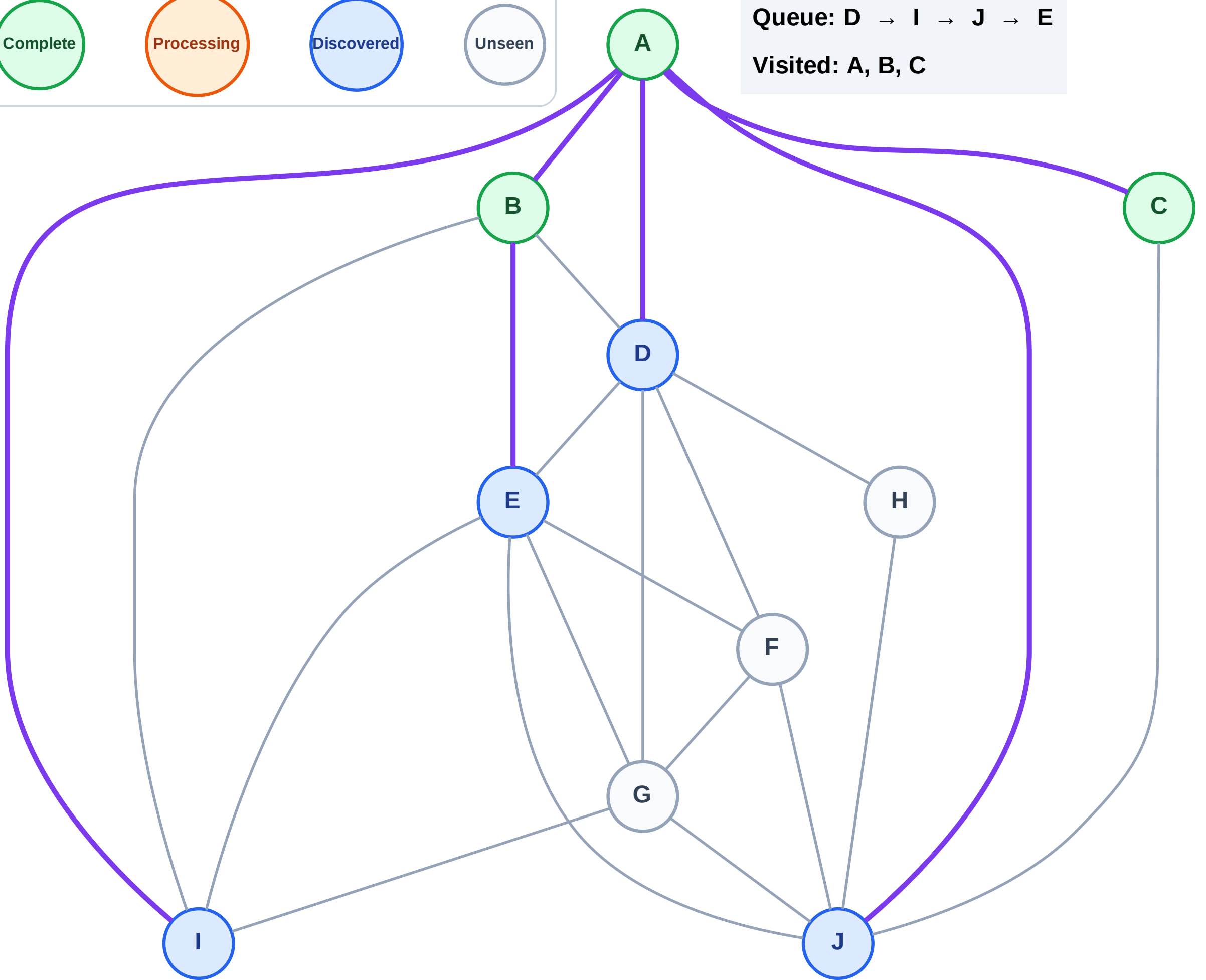
Discovered



Unseen

Queue: D → I → J → E

Visited: A, B, C



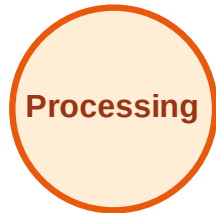
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



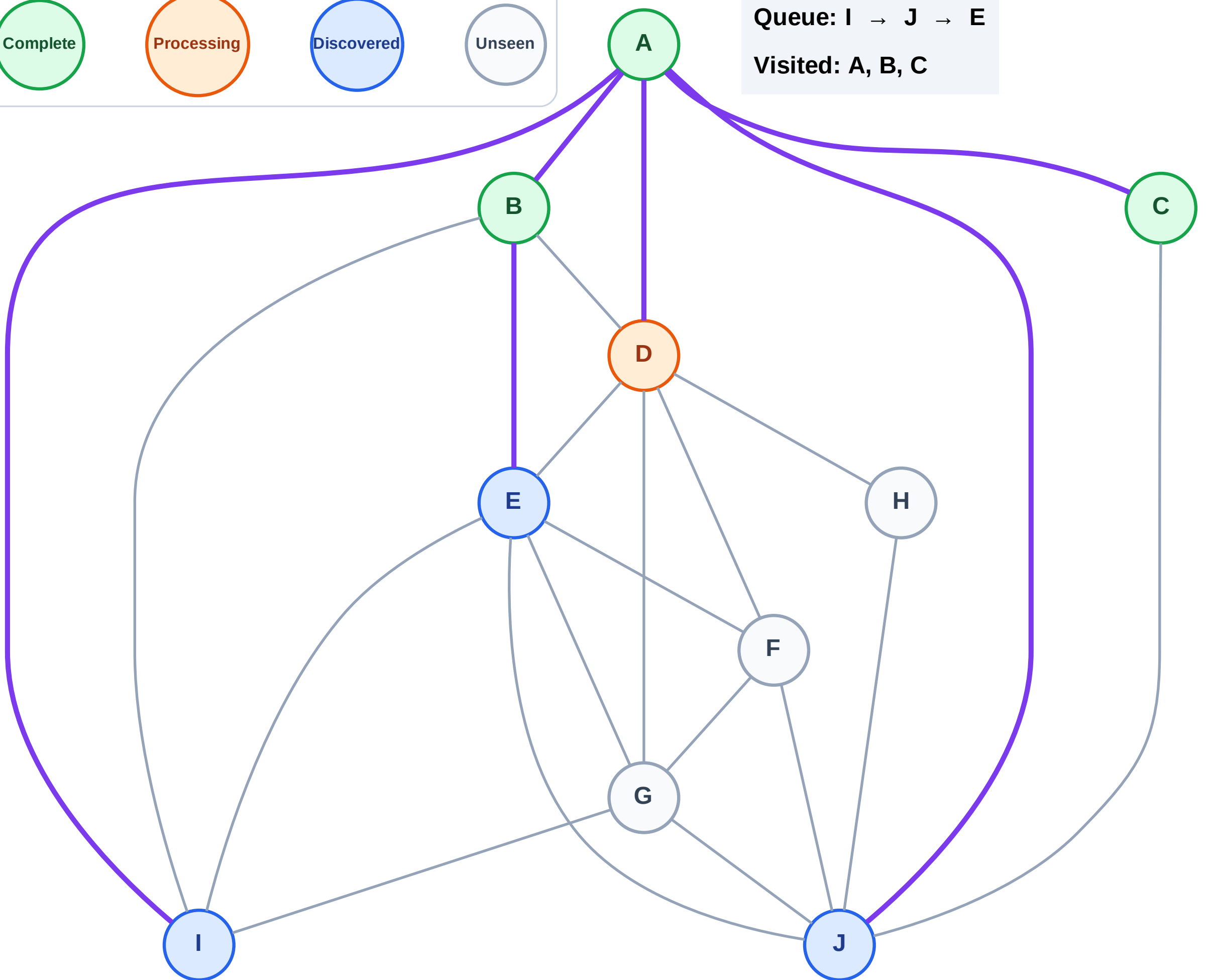
Discovered



Unseen

Queue: I → J → E

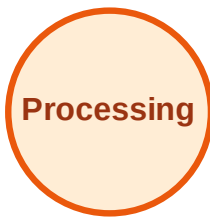
Visited: A, B, C



BFS Traversal

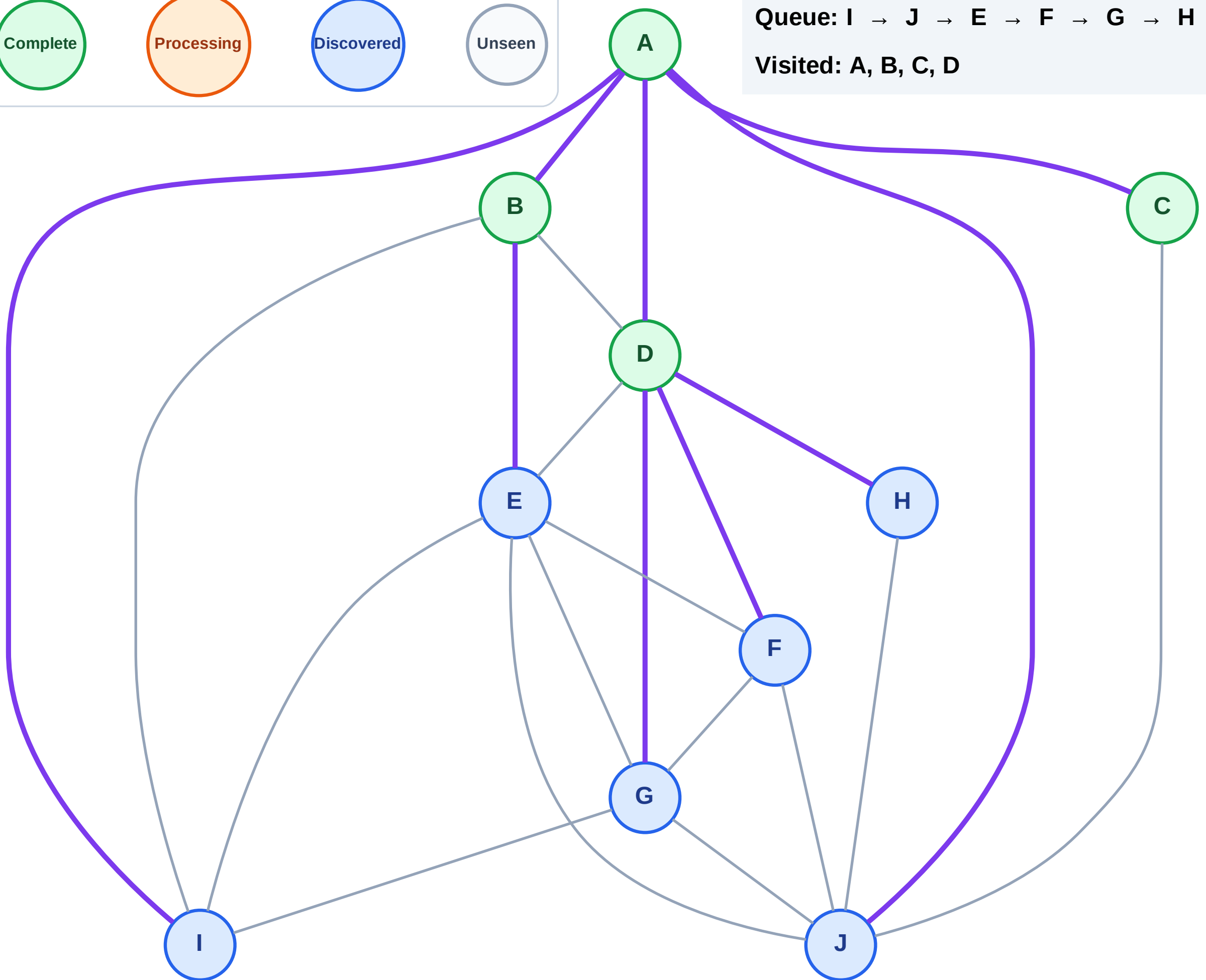
Step 9: Discover F, G, H from D

Legend



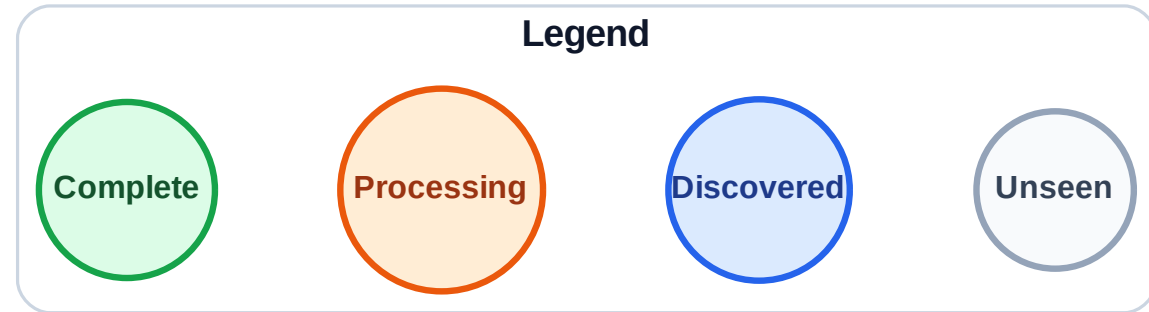
Queue: I → J → E → F → G → H

Visited: A, B, C, D



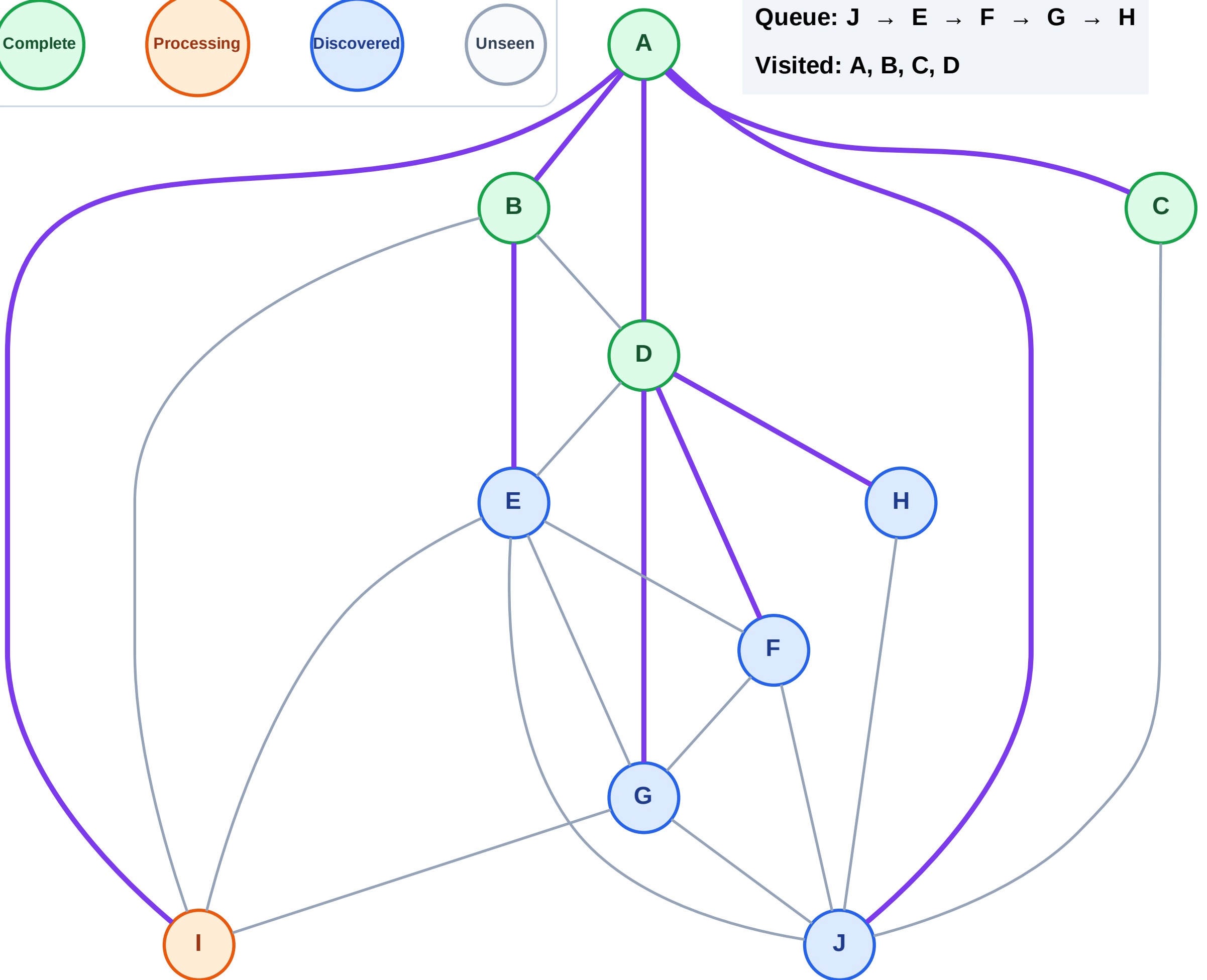
BFS Traversal

Step 10: Process node I



Queue: J → E → F → G → H

Visited: A, B, C, D



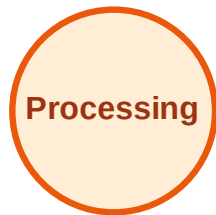
BFS Traversal

Step 11: No new neighbors from I

Legend



Complete



Processing



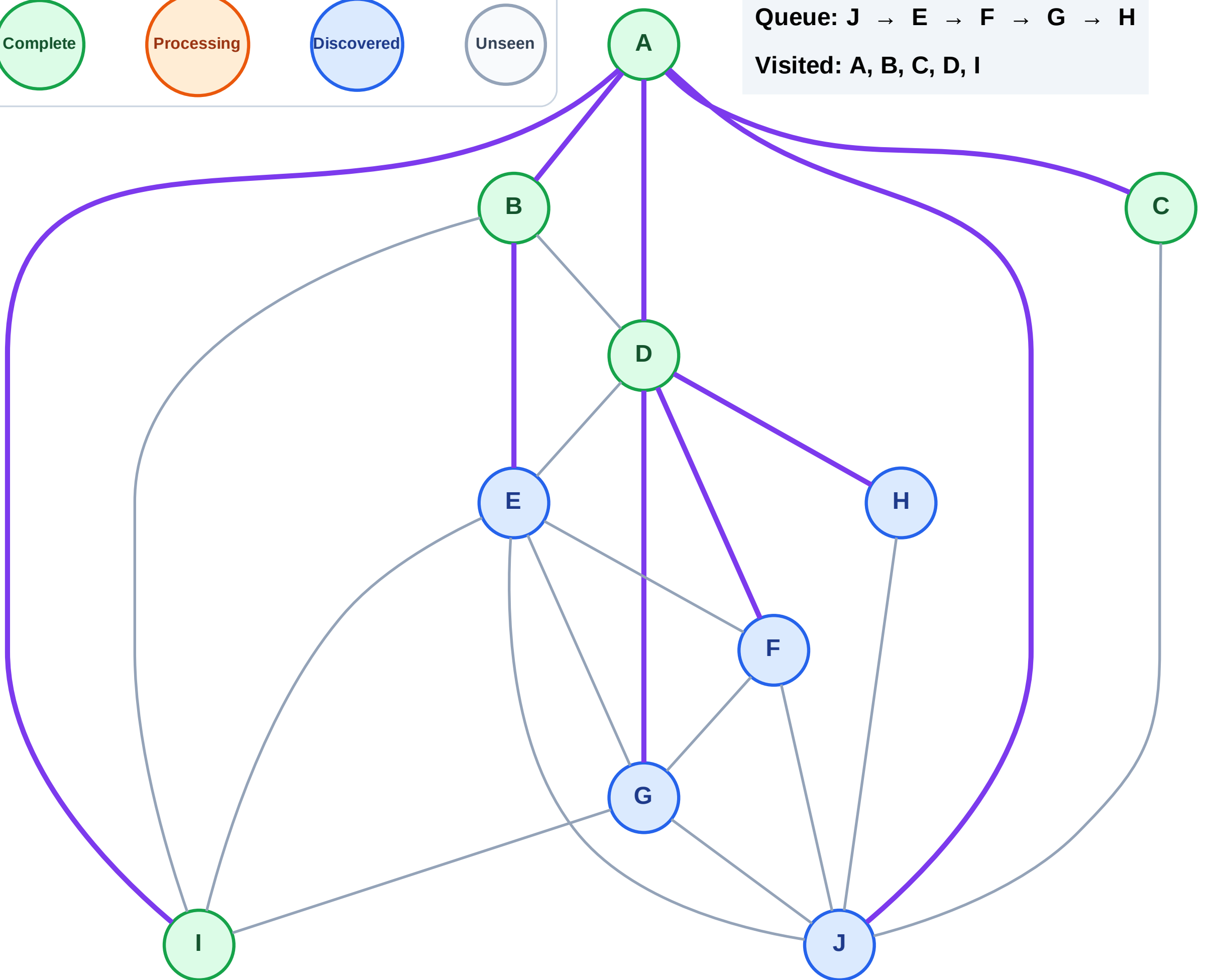
Discovered



Unseen

Queue: J → E → F → G → H

Visited: A, B, C, D, I



BFS Traversal

Step 12: Process node J

Legend

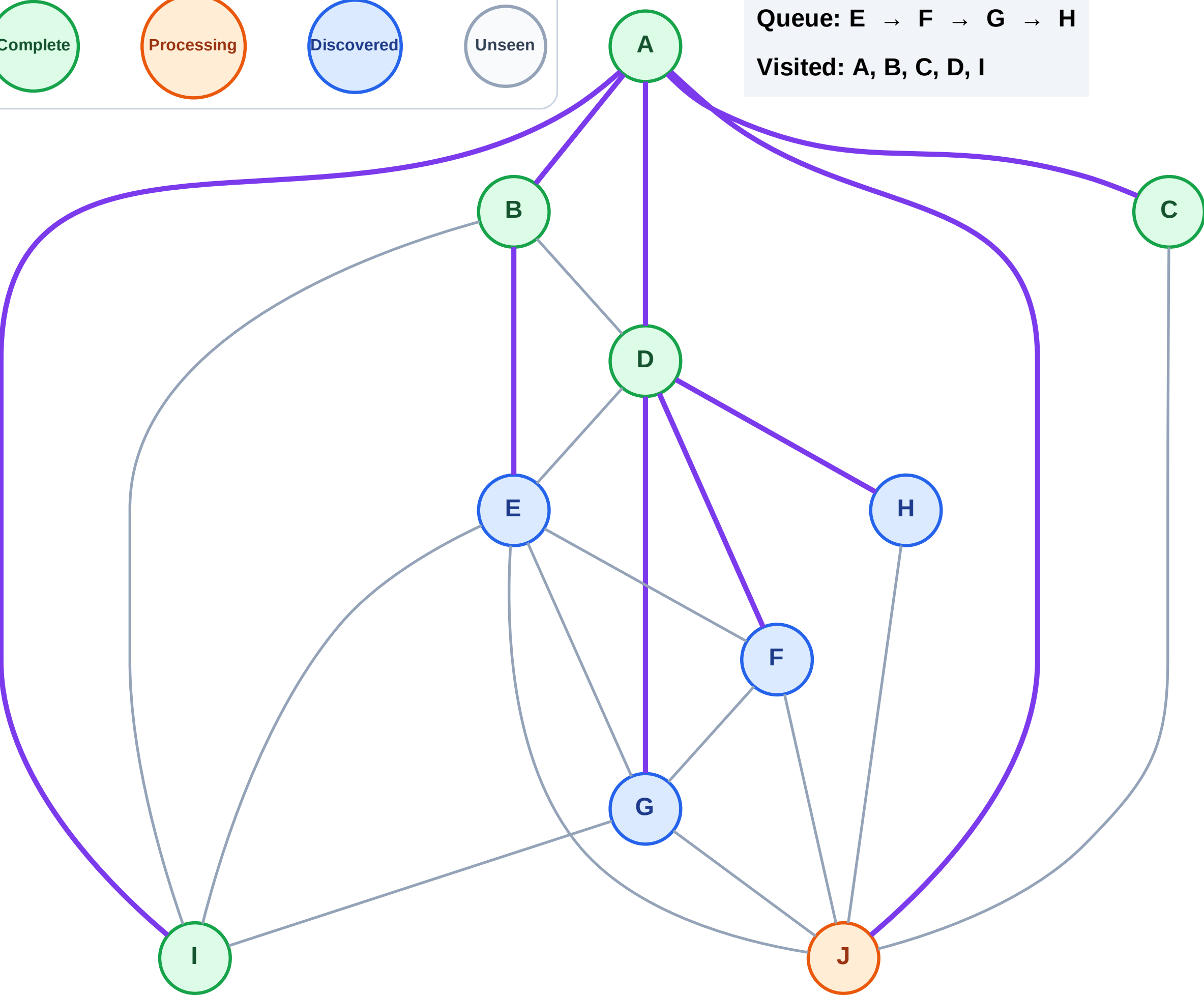
Complete

Processing

Discovered

Unseen

Queue: E → F → G → H
Visited: A, B, C, D, I



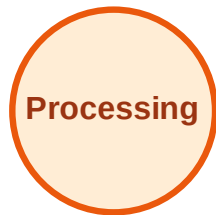
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



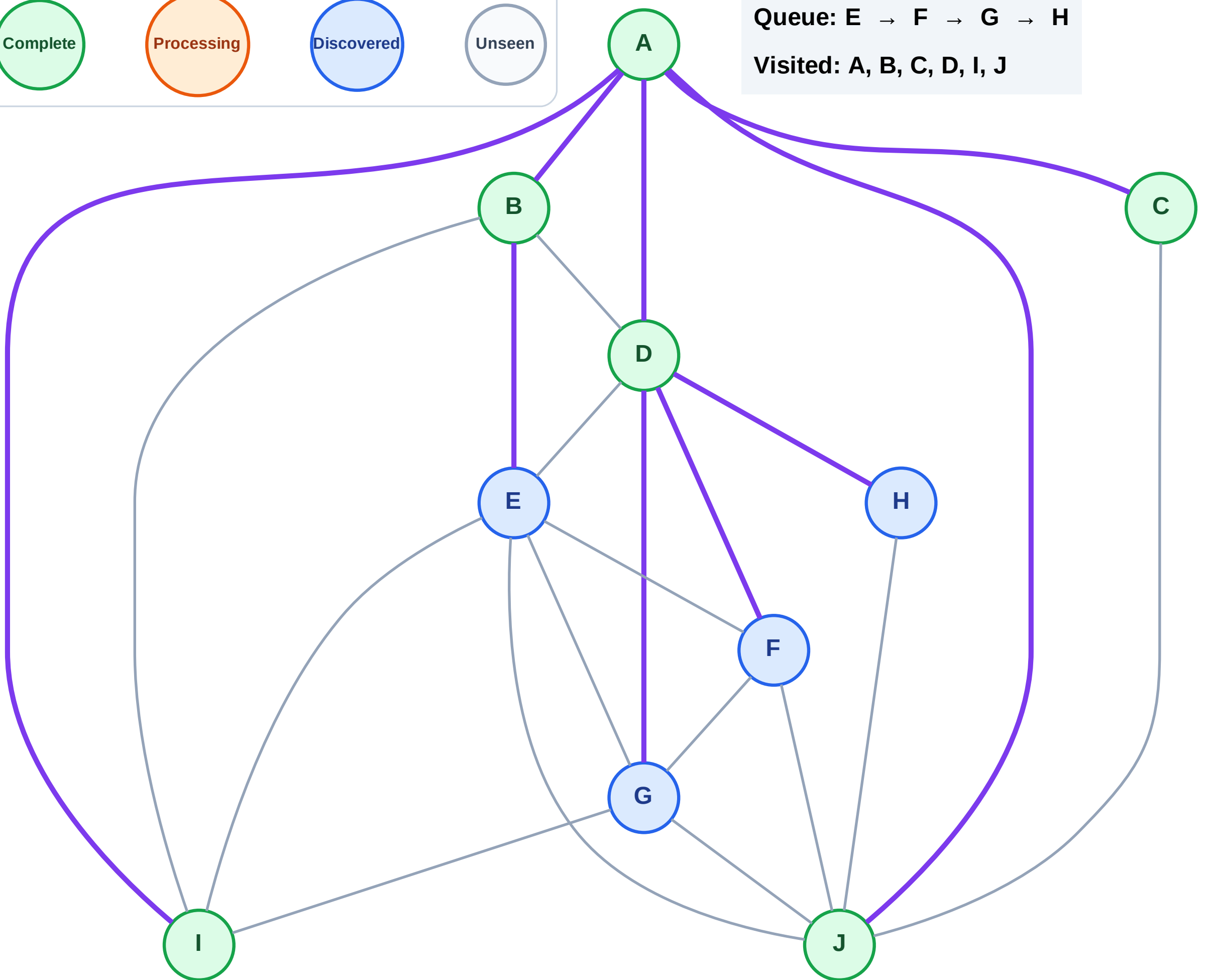
Discovered



Unseen

Queue: E → F → G → H

Visited: A, B, C, D, I, J



BFS Traversal

Step 14: Process node E

Legend

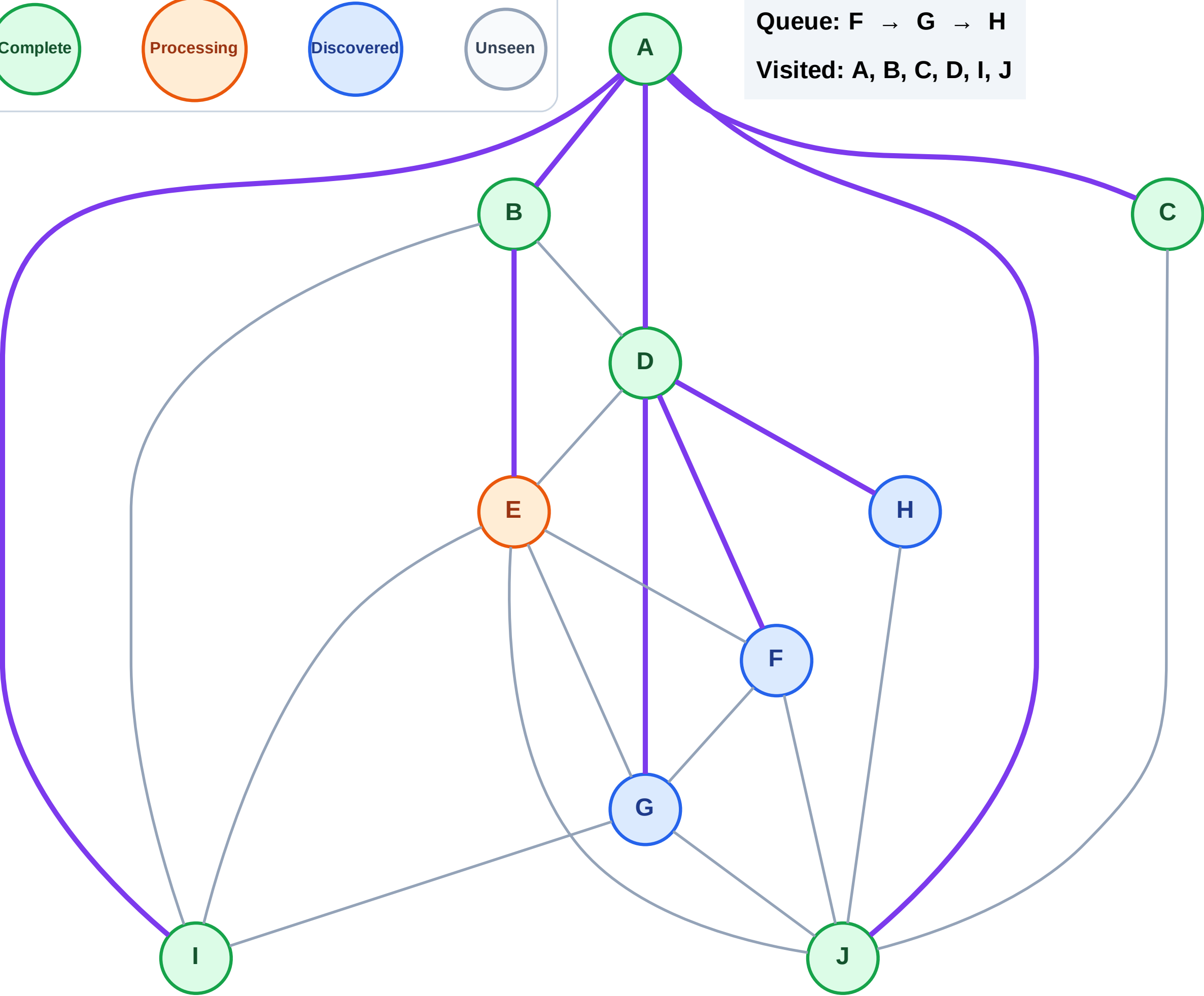
Complete

Processing

Discovered

Unseen

Queue: F → G → H
Visited: A, B, C, D, I, J



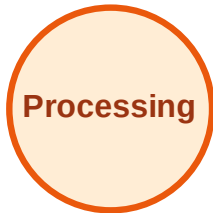
BFS Traversal

Step 15: No new neighbors from E

Legend



Complete



Processing



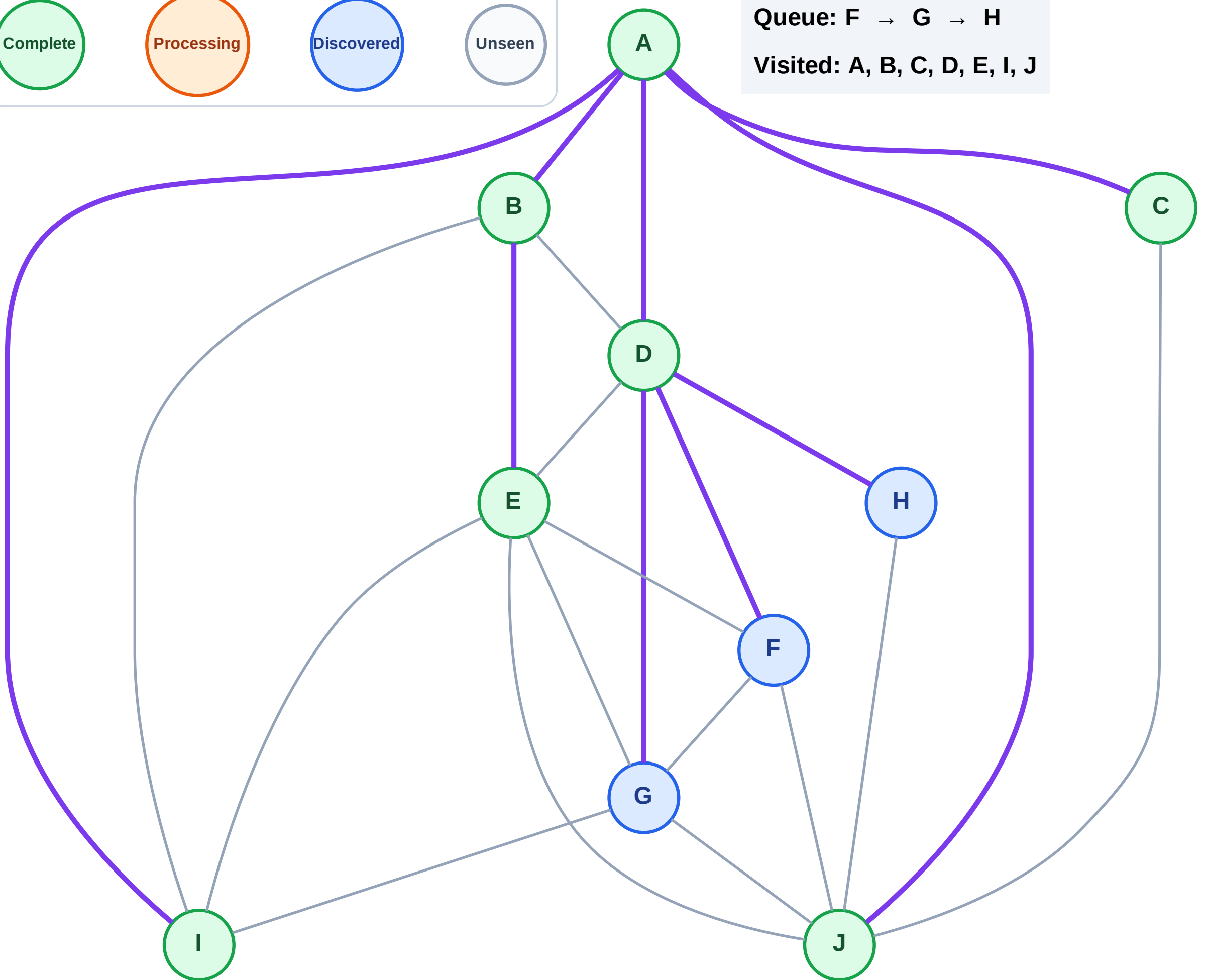
Discovered



Unseen

Queue: F → G → H

Visited: A, B, C, D, E, I, J



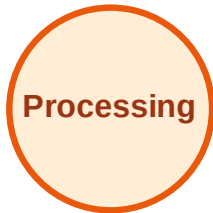
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



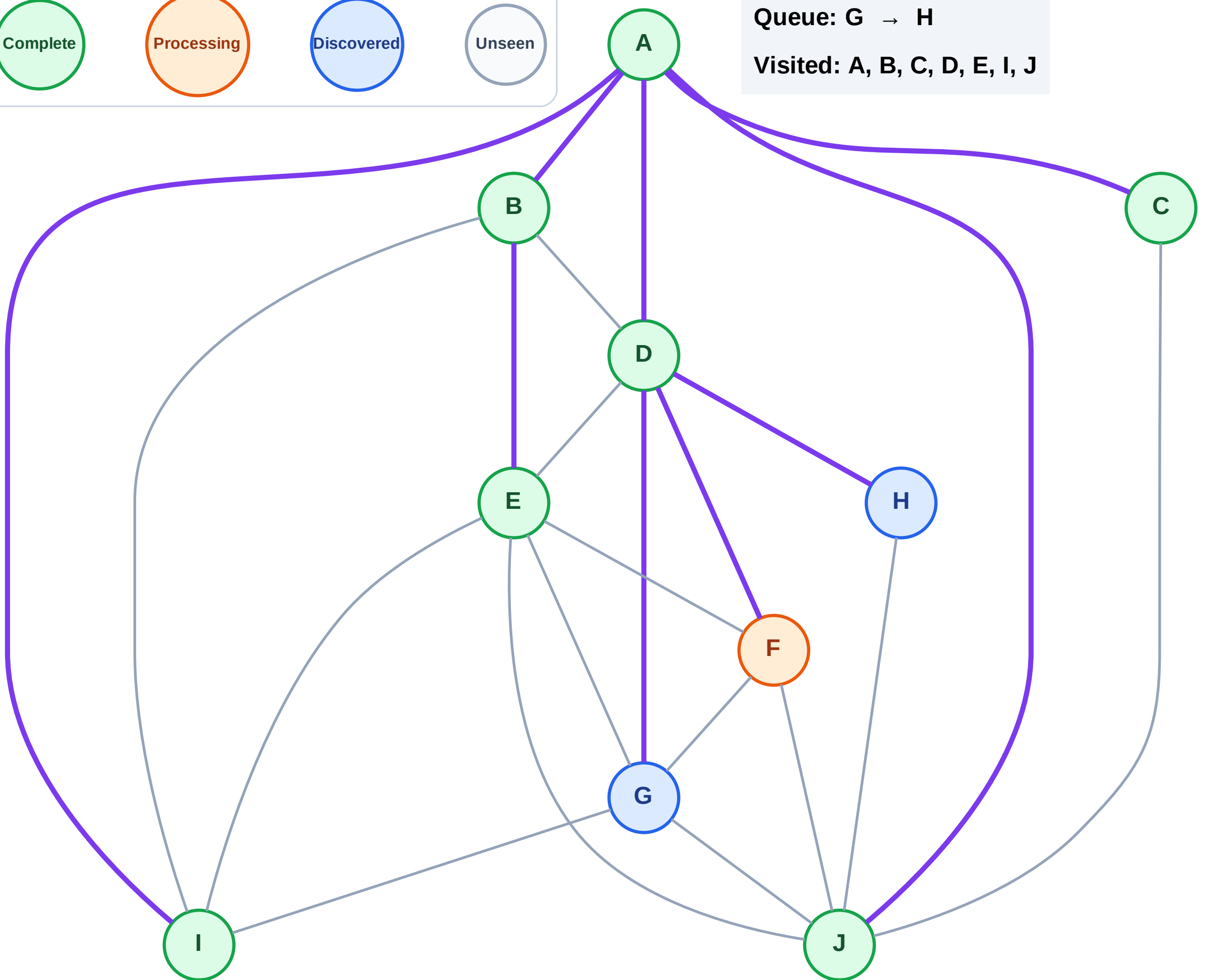
Discovered



Unseen

Queue: G → H

Visited: A, B, C, D, E, I, J



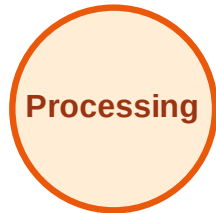
BFS Traversal

Step 17: No new neighbors from F

Legend



Complete



Processing



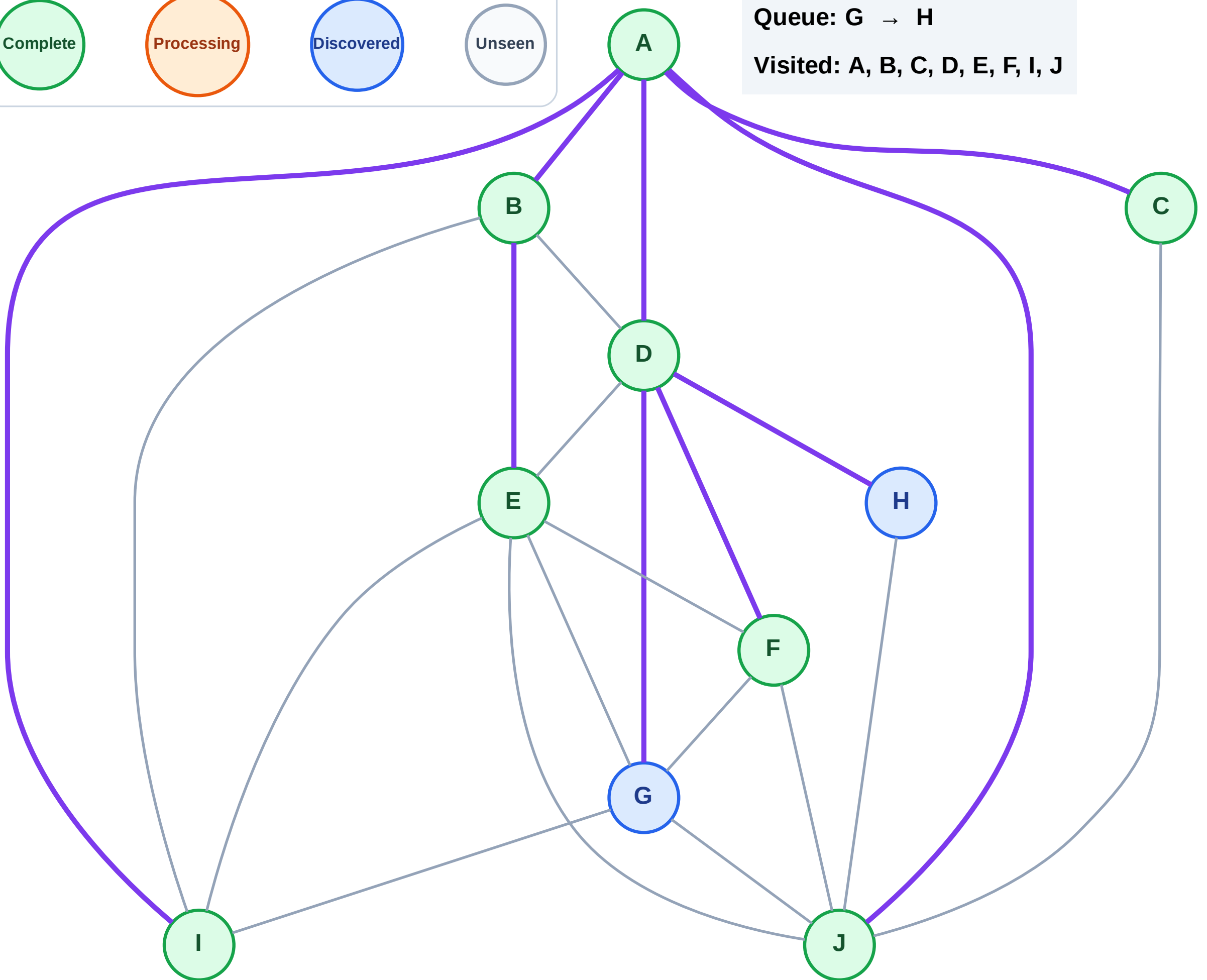
Discovered



Unseen

Queue: G → H

Visited: A, B, C, D, E, F, I, J



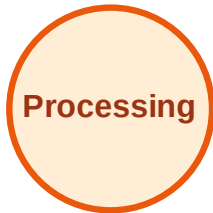
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



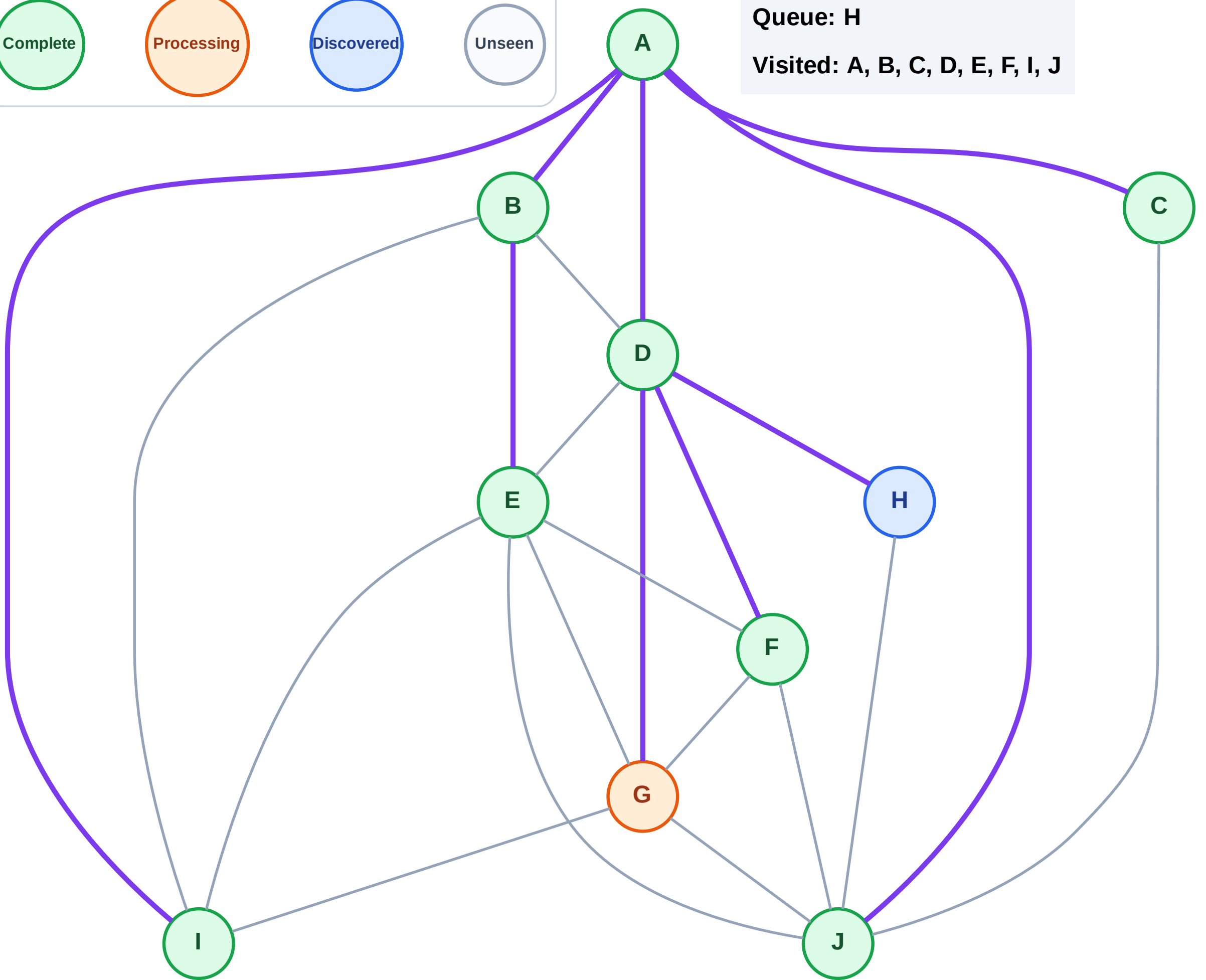
Discovered



Unseen

Queue: H

Visited: A, B, C, D, E, F, I, J



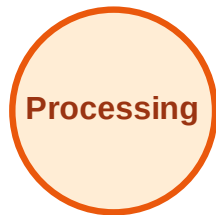
BFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



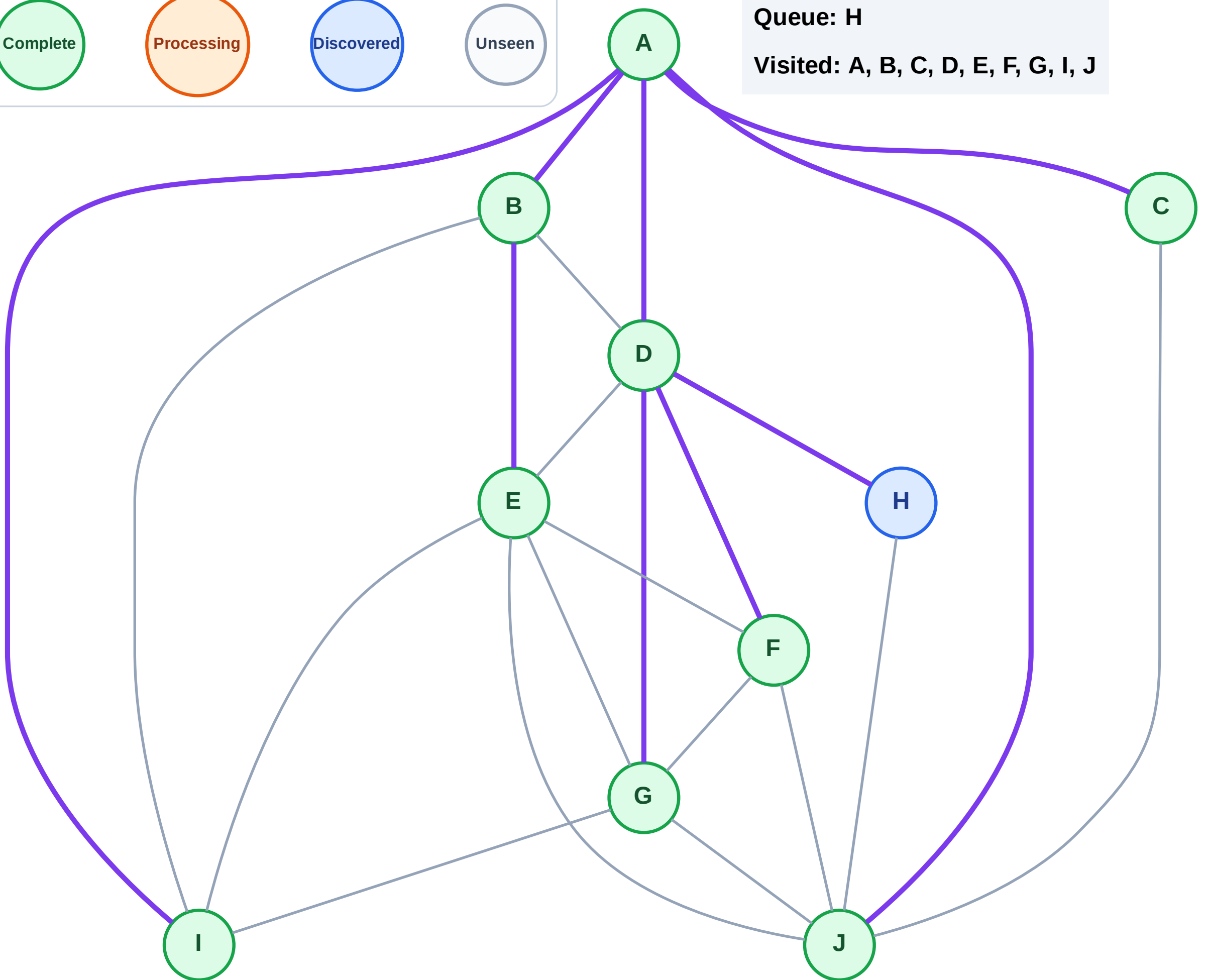
Discovered



Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I, J



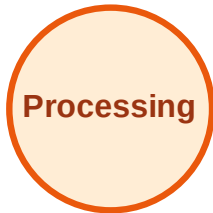
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



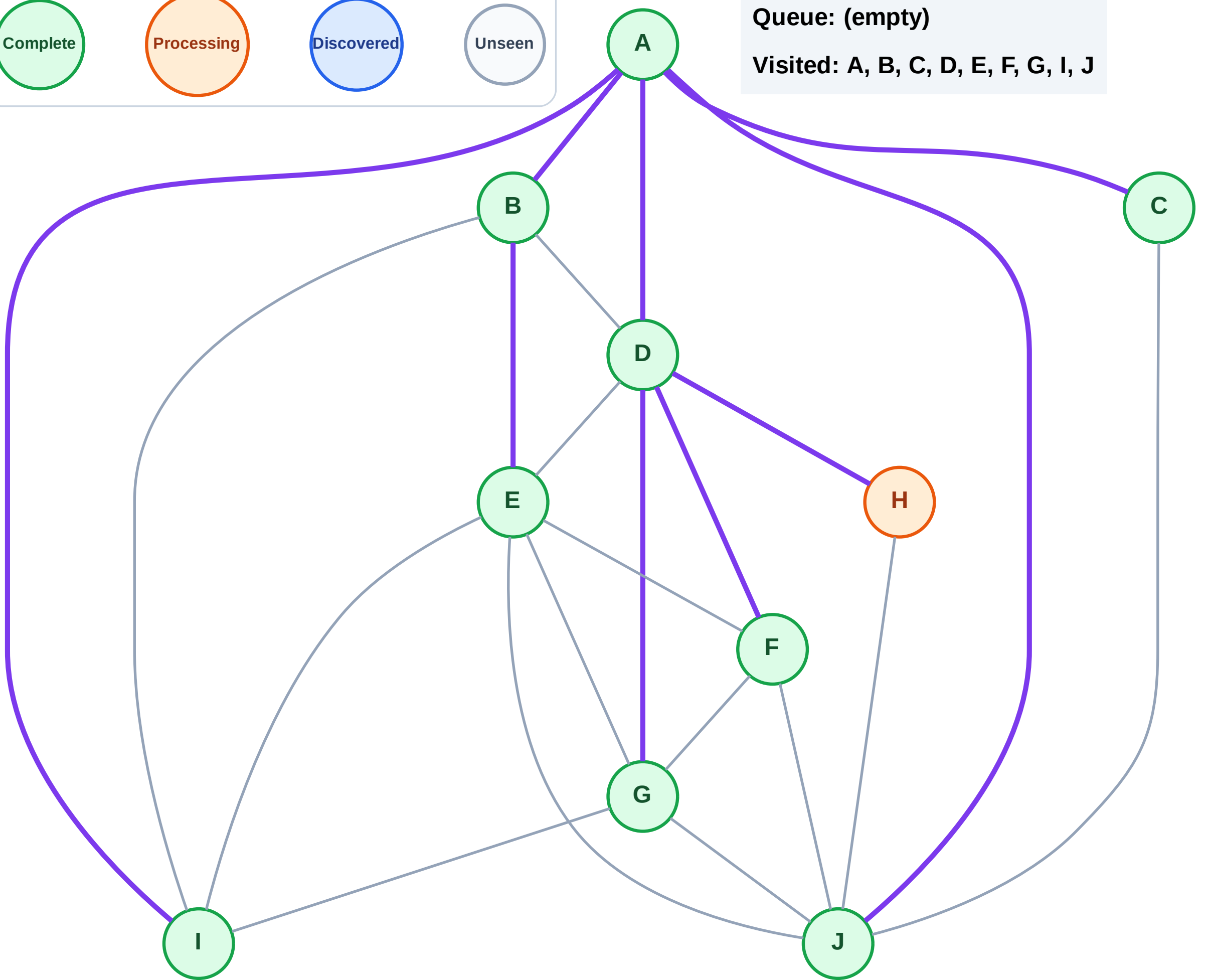
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



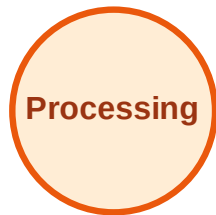
BFS Traversal

Step 21: No new neighbors from H

Legend



Complete



Processing



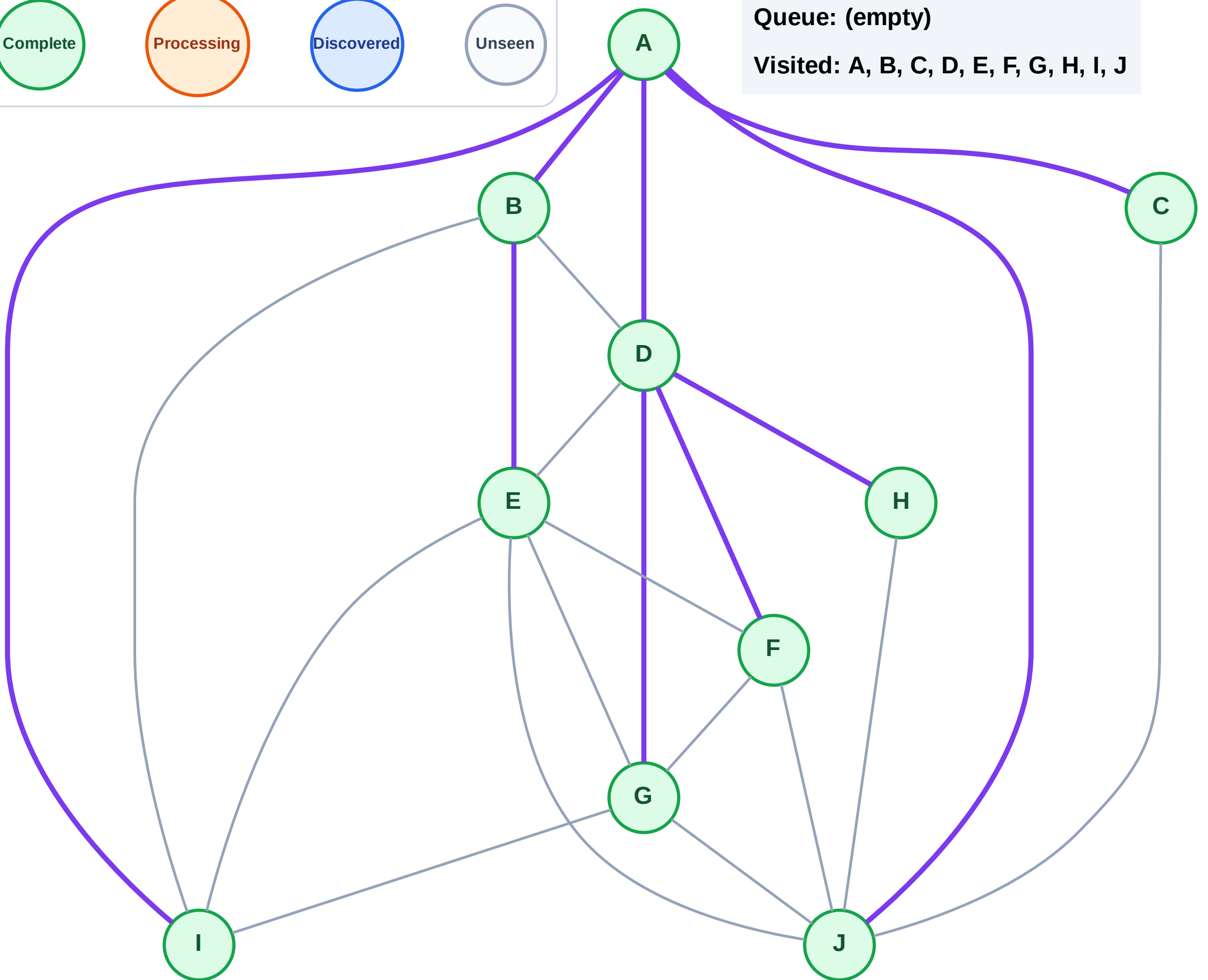
Discovered



Unseen

Queue: (empty)

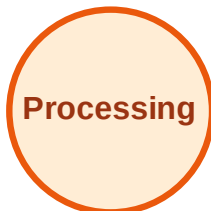
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

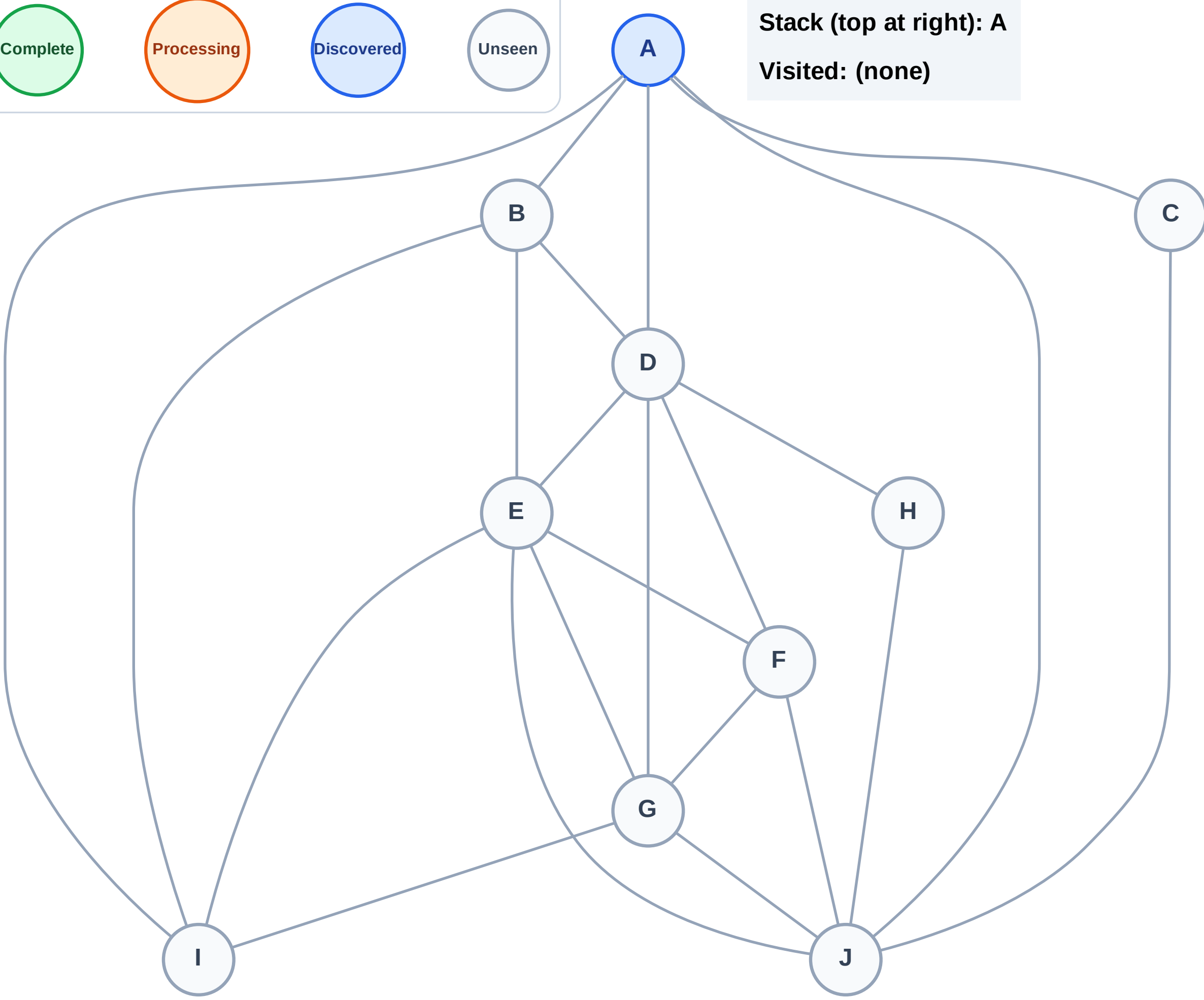
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



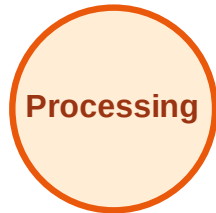
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



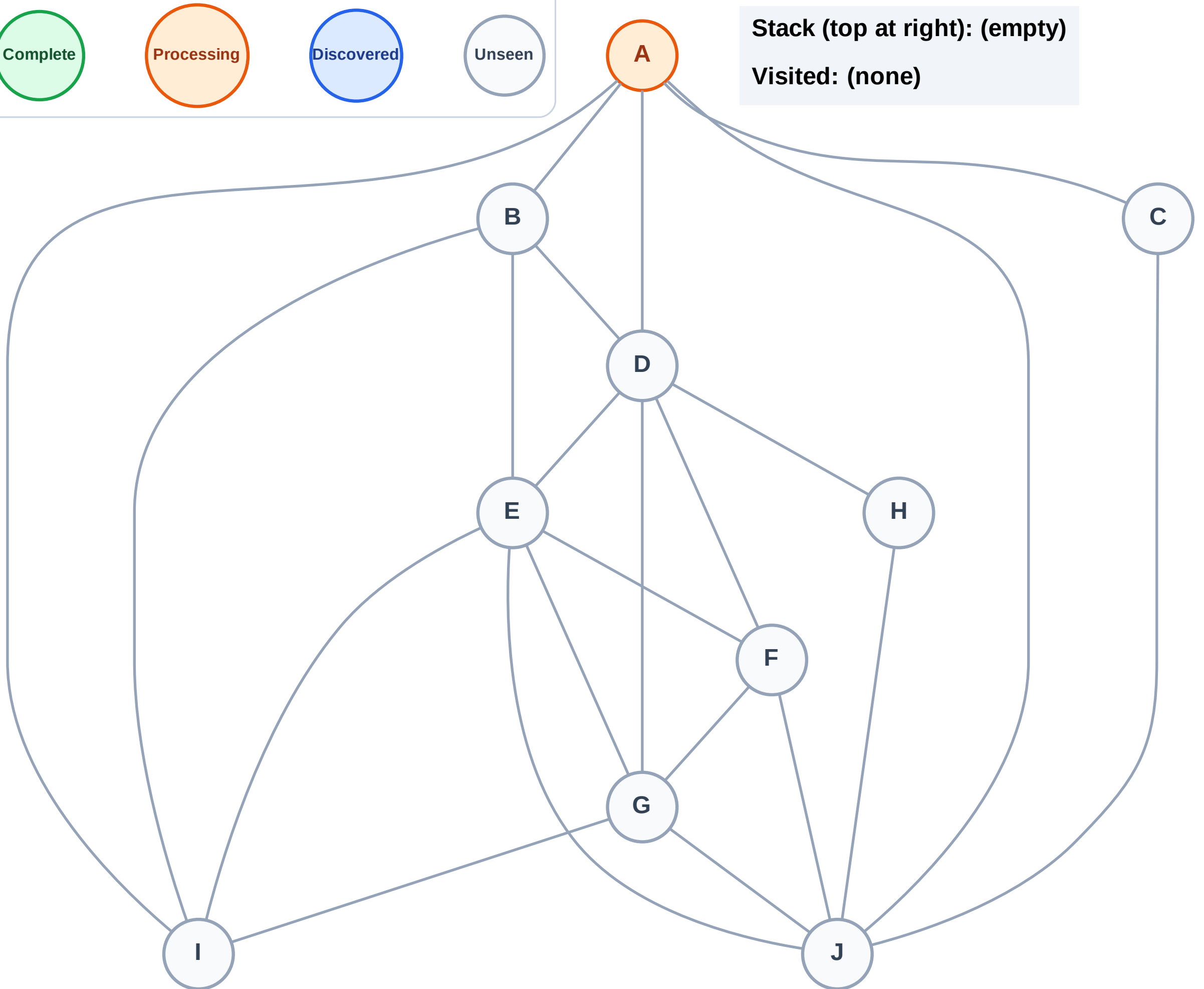
Discovered



Unseen

Stack (top at right): (empty)

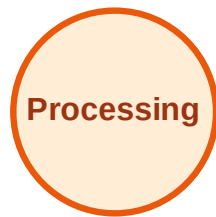
Visited: (none)



DFS Traversal

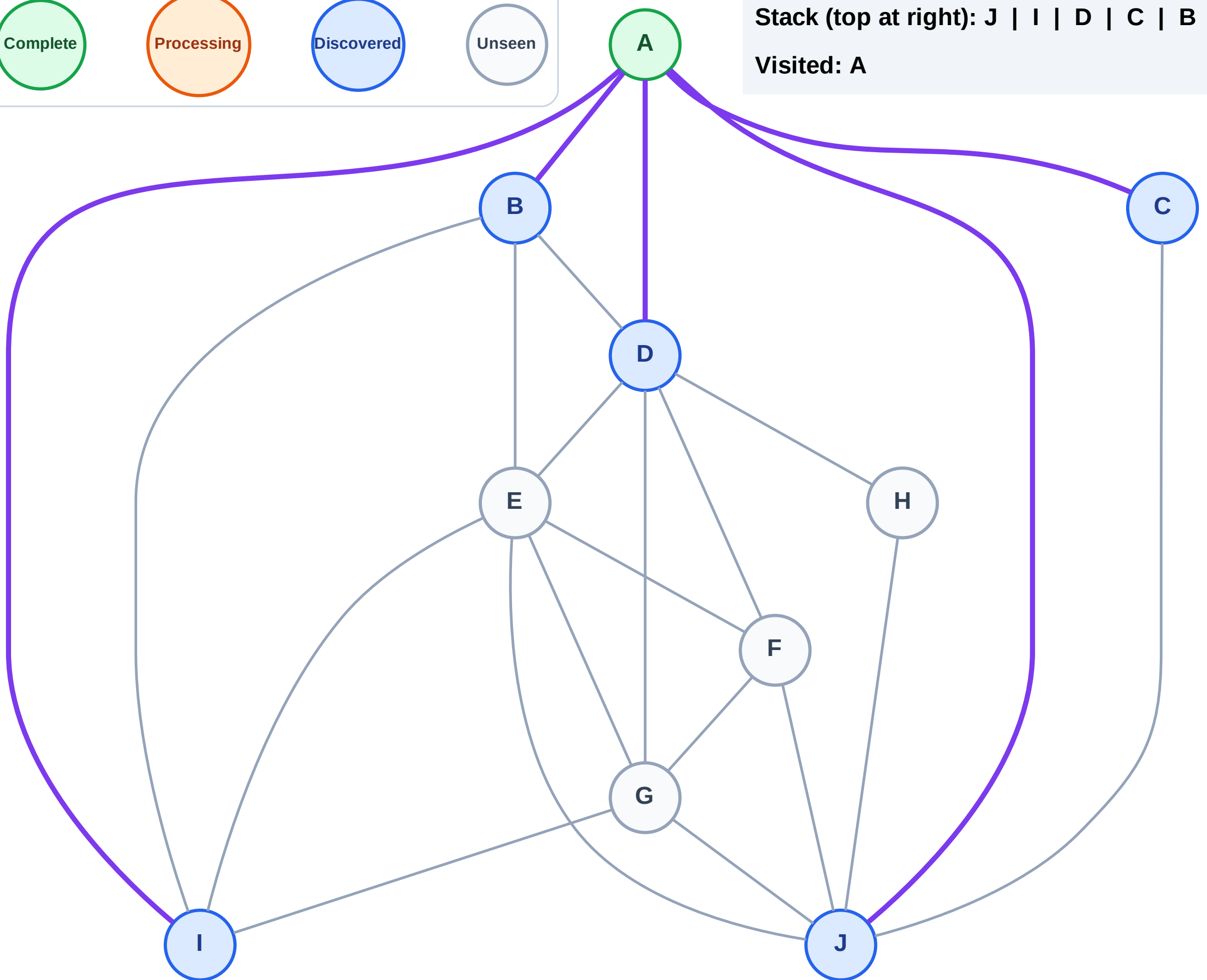
Step 3: Discover J, I, D, C, B from A

Legend



Stack (top at right): J | I | D | C | B

Visited: A



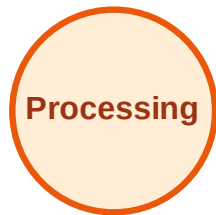
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



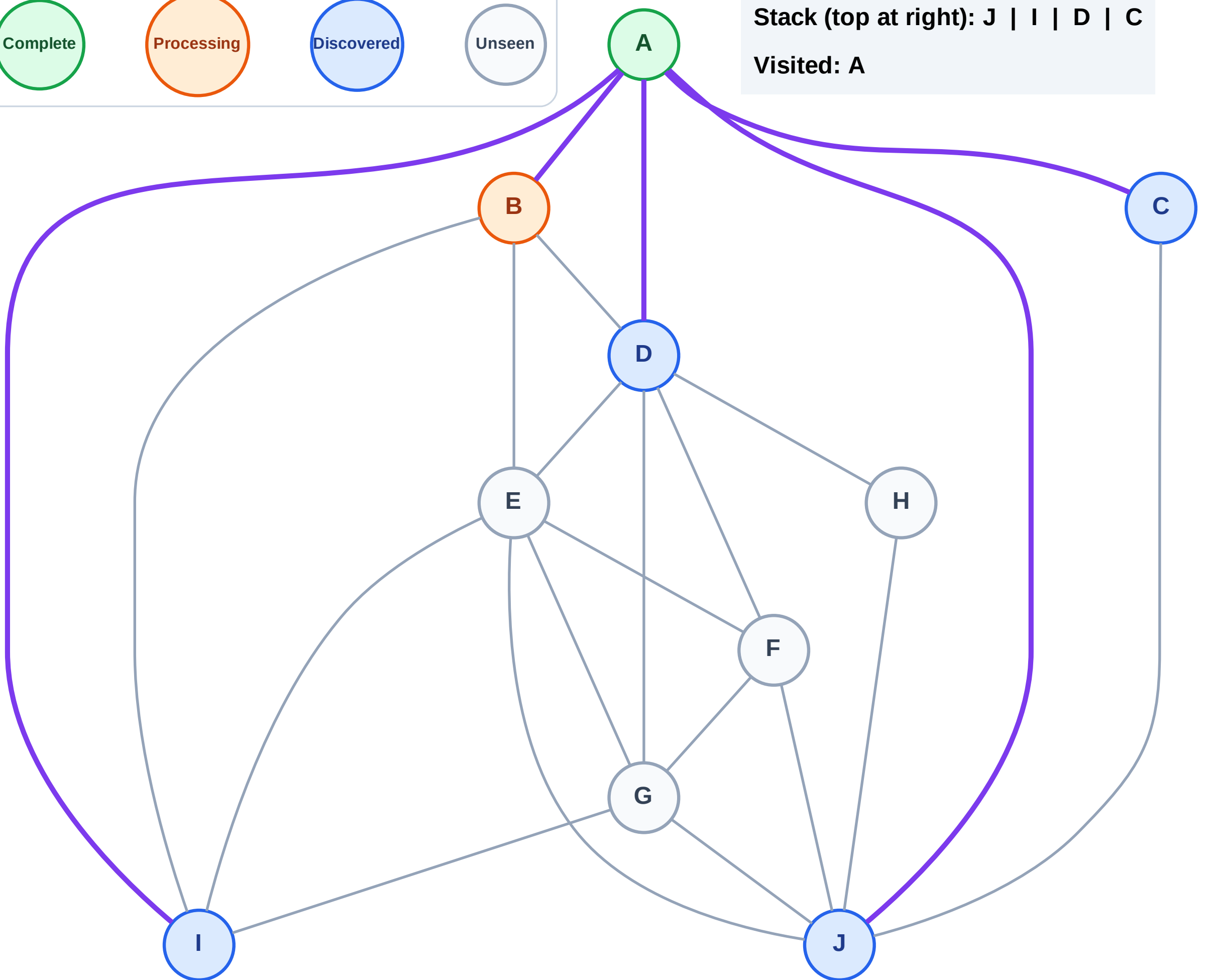
Discovered



Unseen

Stack (top at right): J | I | D | C

Visited: A

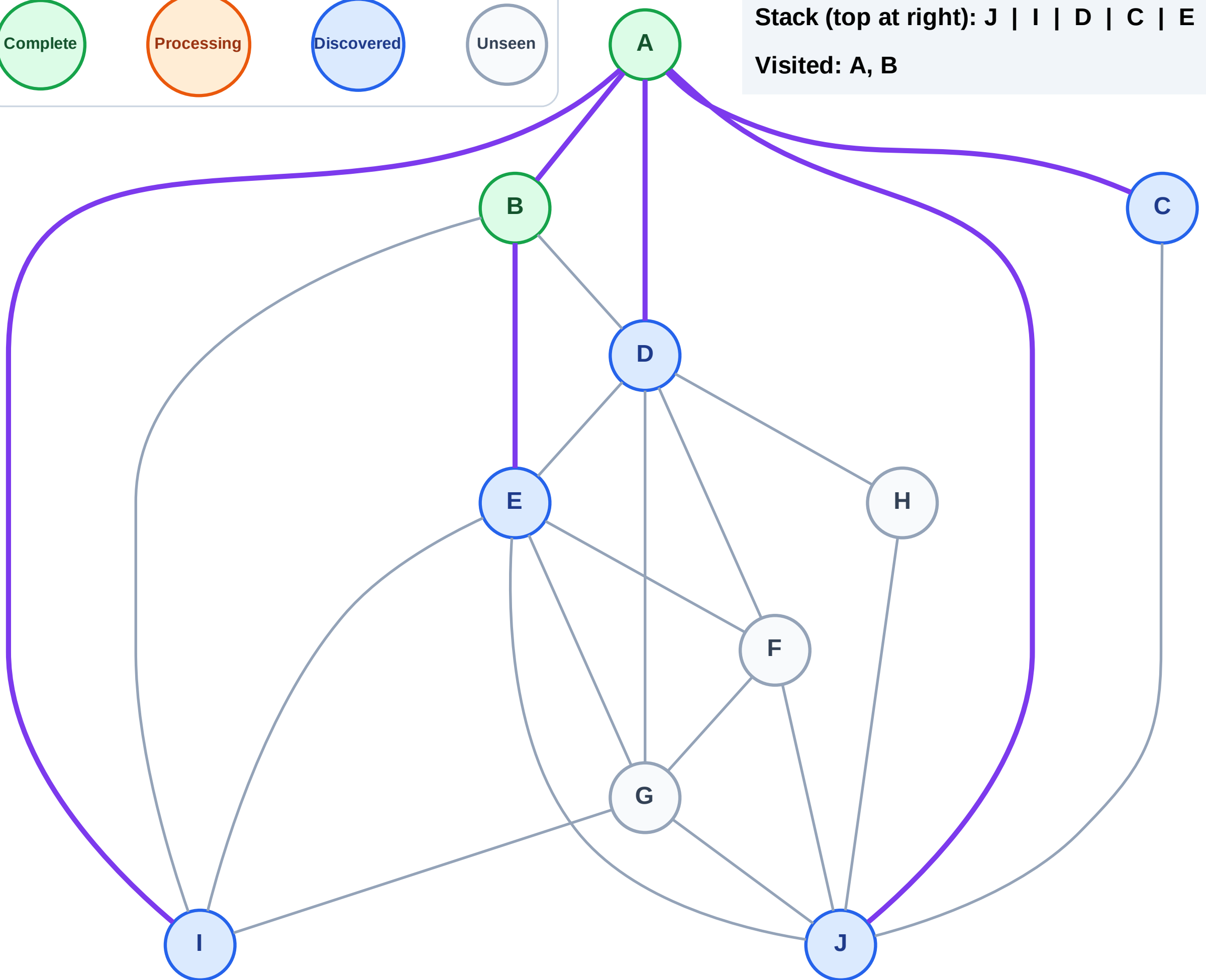
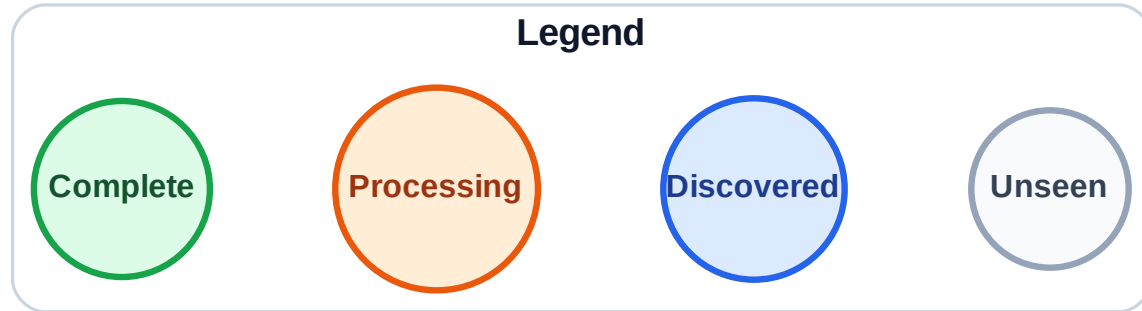


DFS Traversal

Step 5: Discover E from B

Stack (top at right): J | I | D | C | E

Visited: A, B



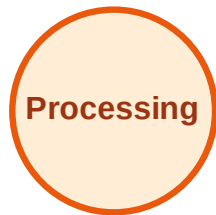
DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



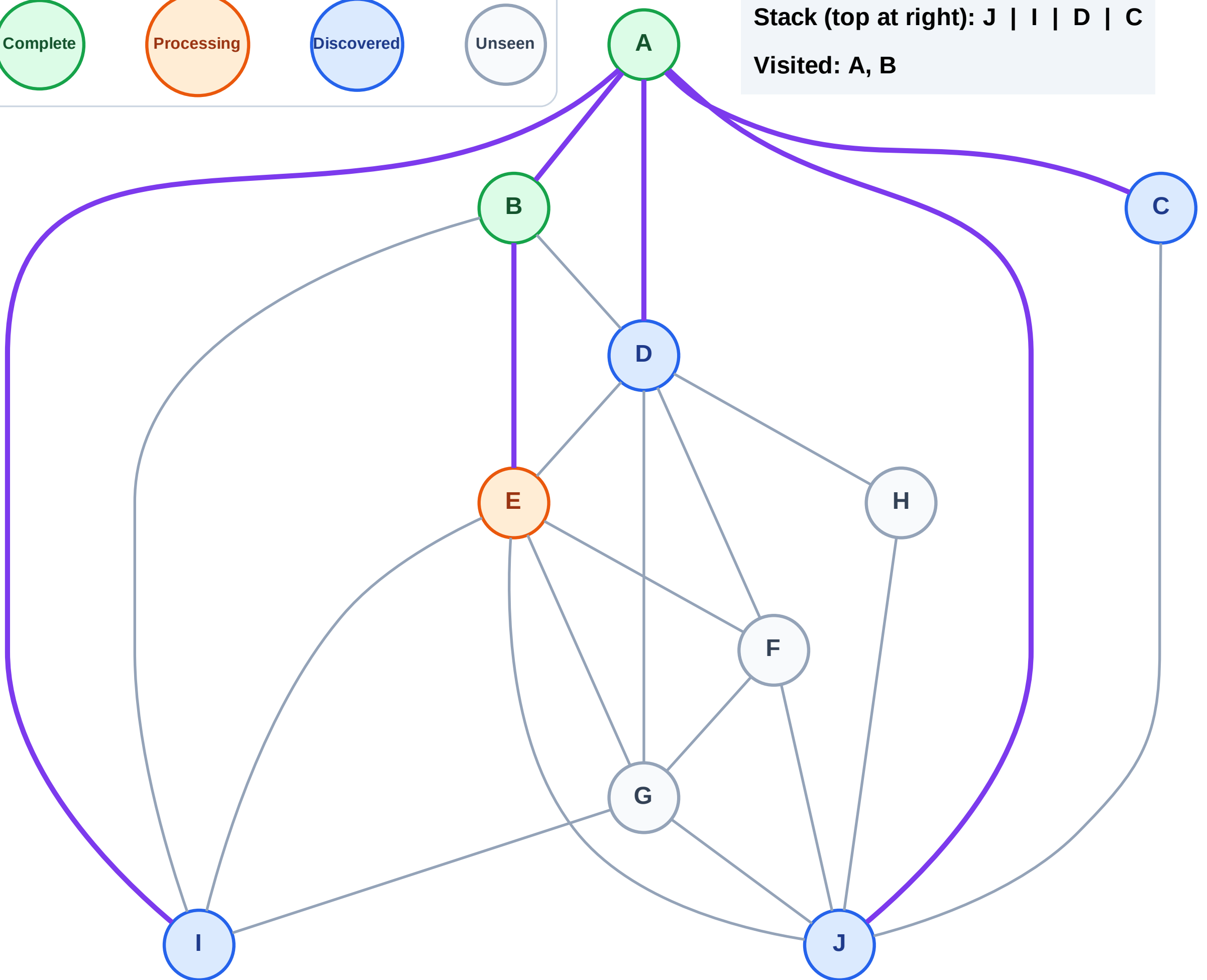
Discovered



Unseen

Stack (top at right): J | I | D | C

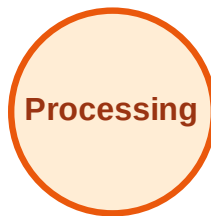
Visited: A, B



DFS Traversal

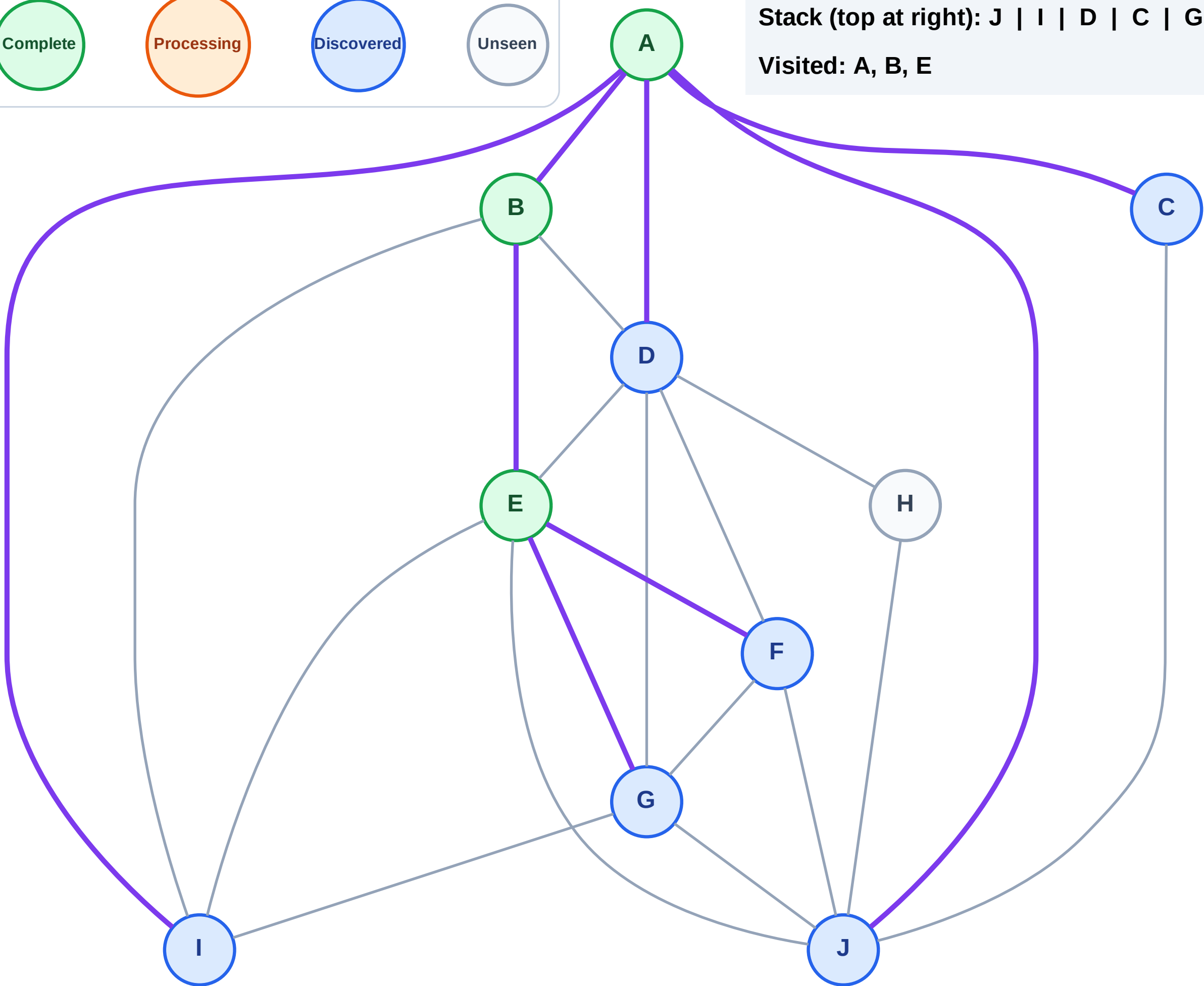
Step 7: Discover G, F from E

Legend



Stack (top at right): J | I | D | C | G | F

Visited: A, B, E



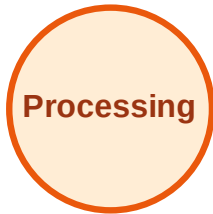
DFS Traversal

Step 8: Process node F

Legend



Complete



Processing



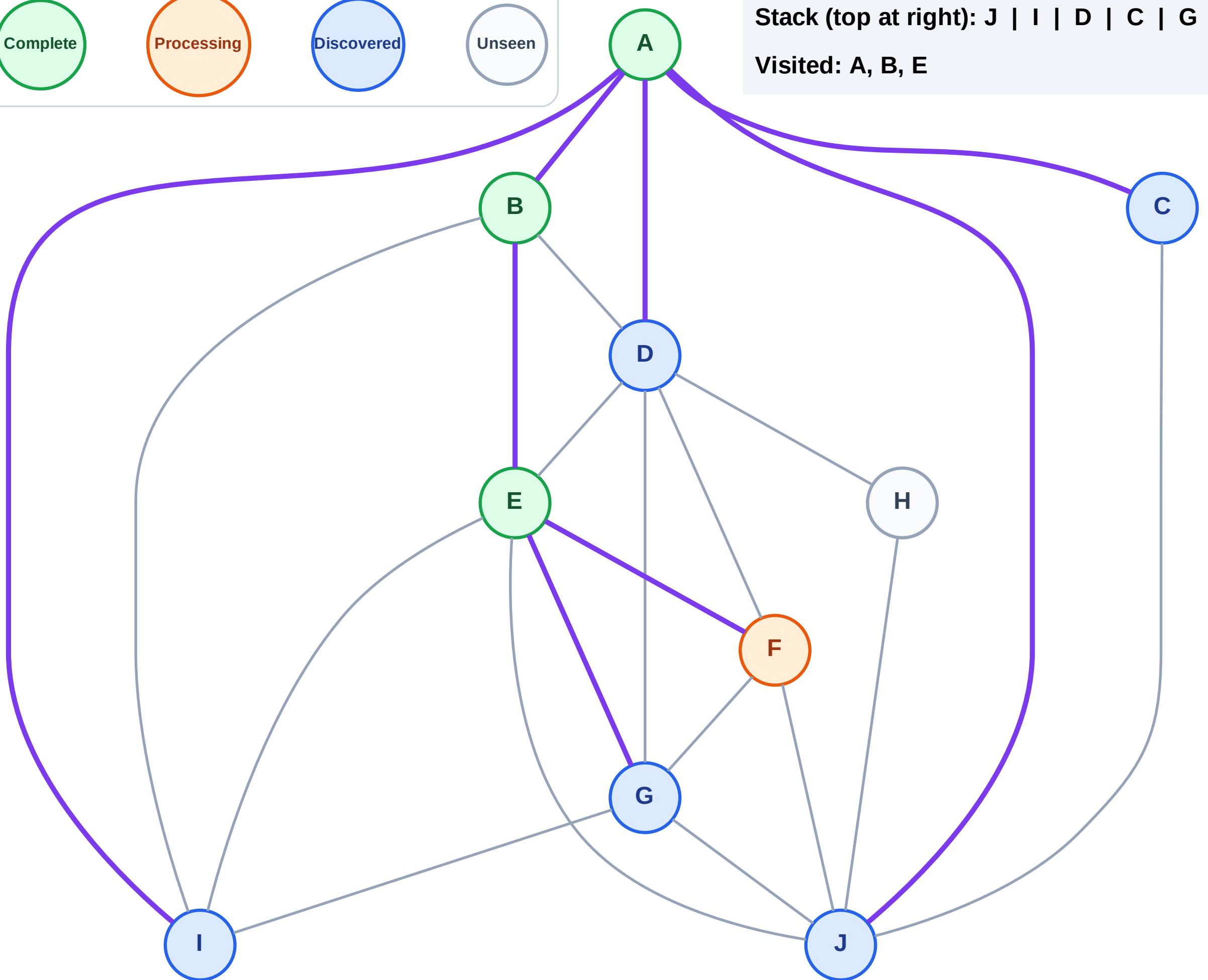
Discovered



Unseen

Stack (top at right): J | I | D | C | G

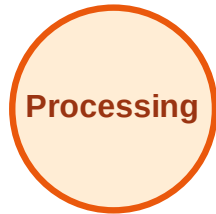
Visited: A, B, E



DFS Traversal

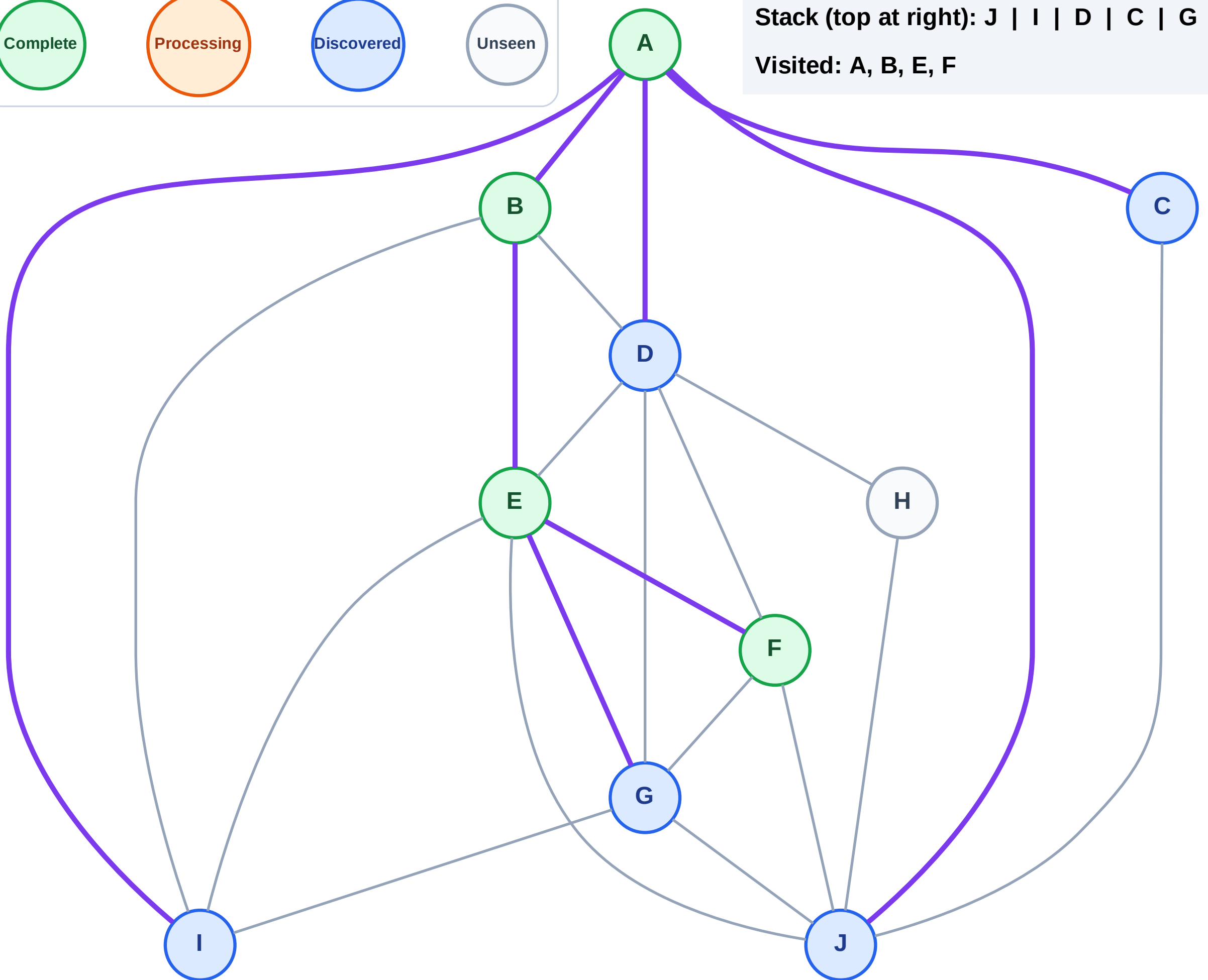
Step 9: No new neighbors from F

Legend



Stack (top at right): J | I | D | C | G

Visited: A, B, E, F



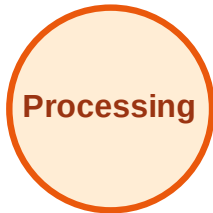
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



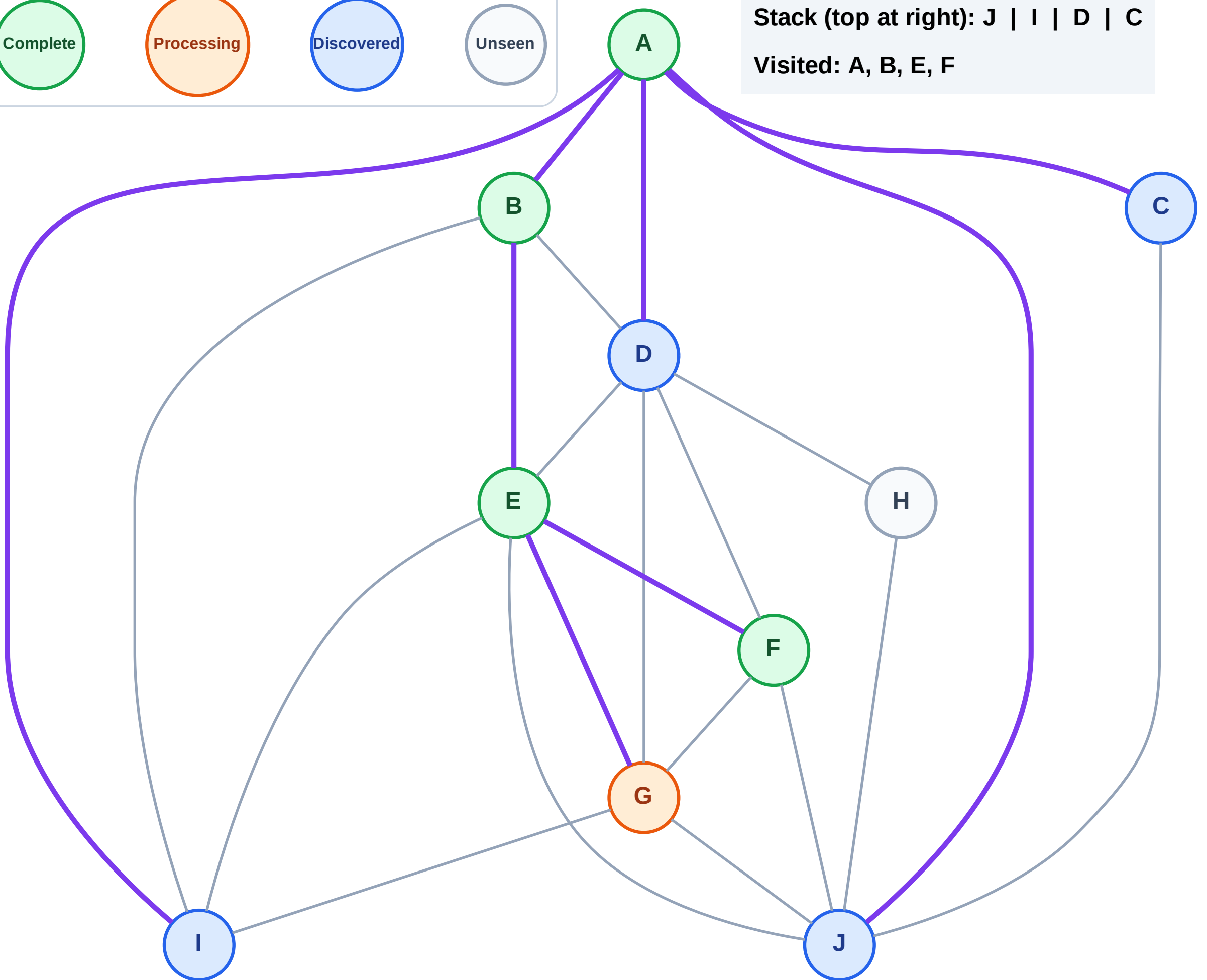
Discovered



Unseen

Stack (top at right): J | I | D | C

Visited: A, B, E, F



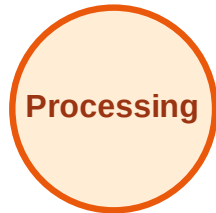
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



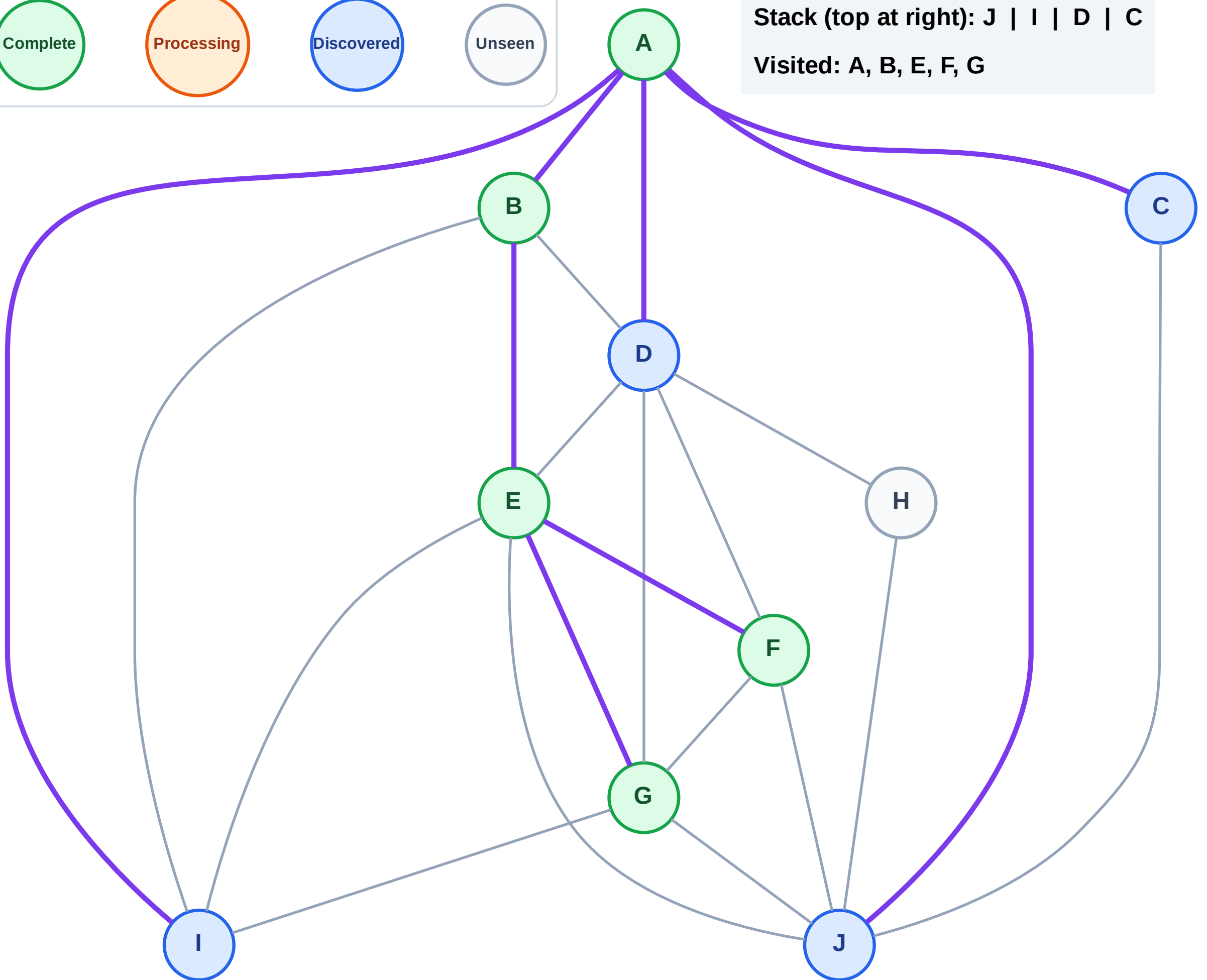
Discovered



Unseen

Stack (top at right): J | I | D | C

Visited: A, B, E, F, G



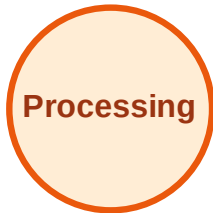
DFS Traversal

Step 12: Process node C

Legend



Complete



Processing



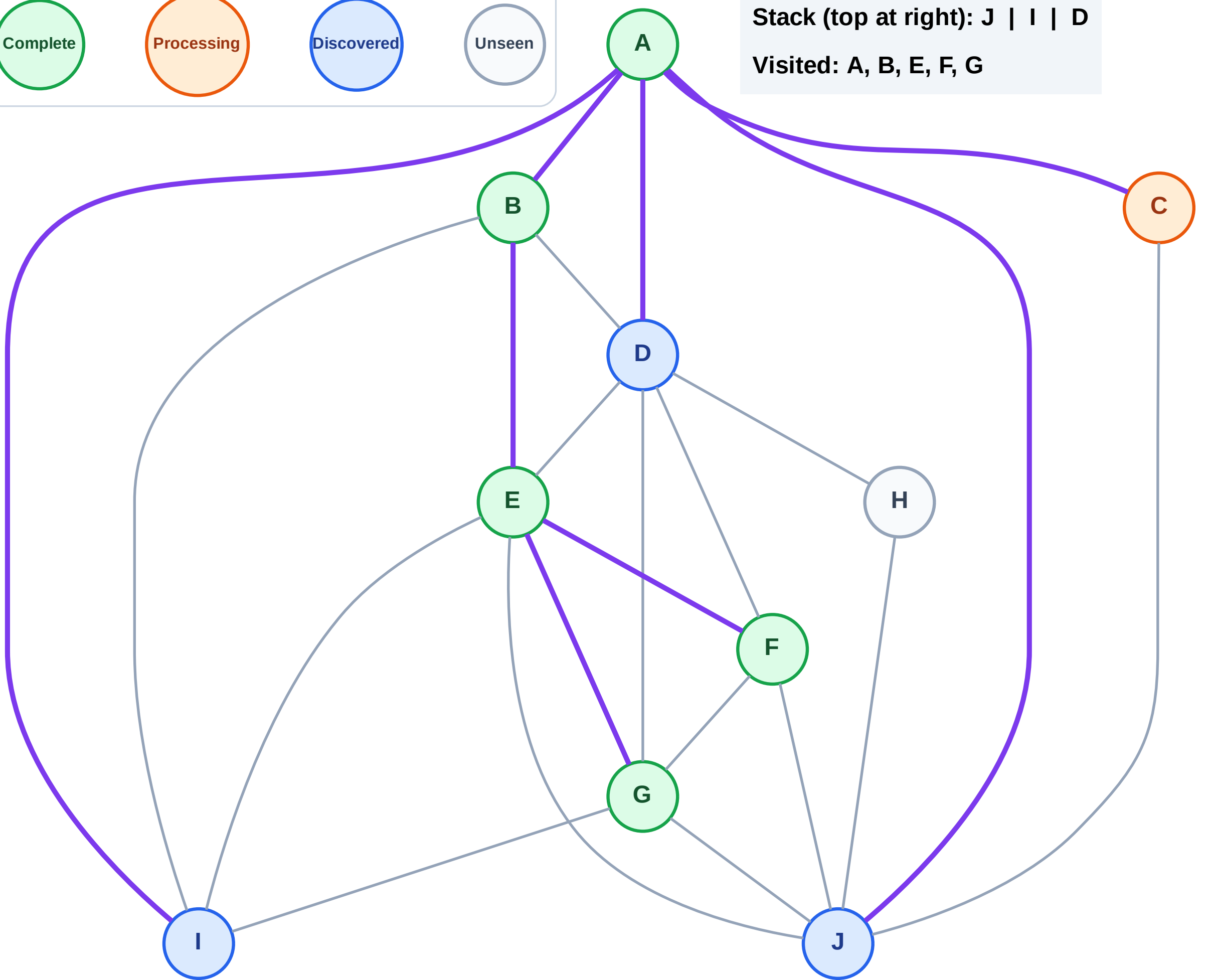
Discovered



Unseen

Stack (top at right): J | I | D

Visited: A, B, E, F, G



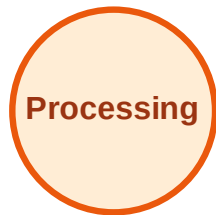
DFS Traversal

Step 13: No new neighbors from C

Legend



Complete



Processing



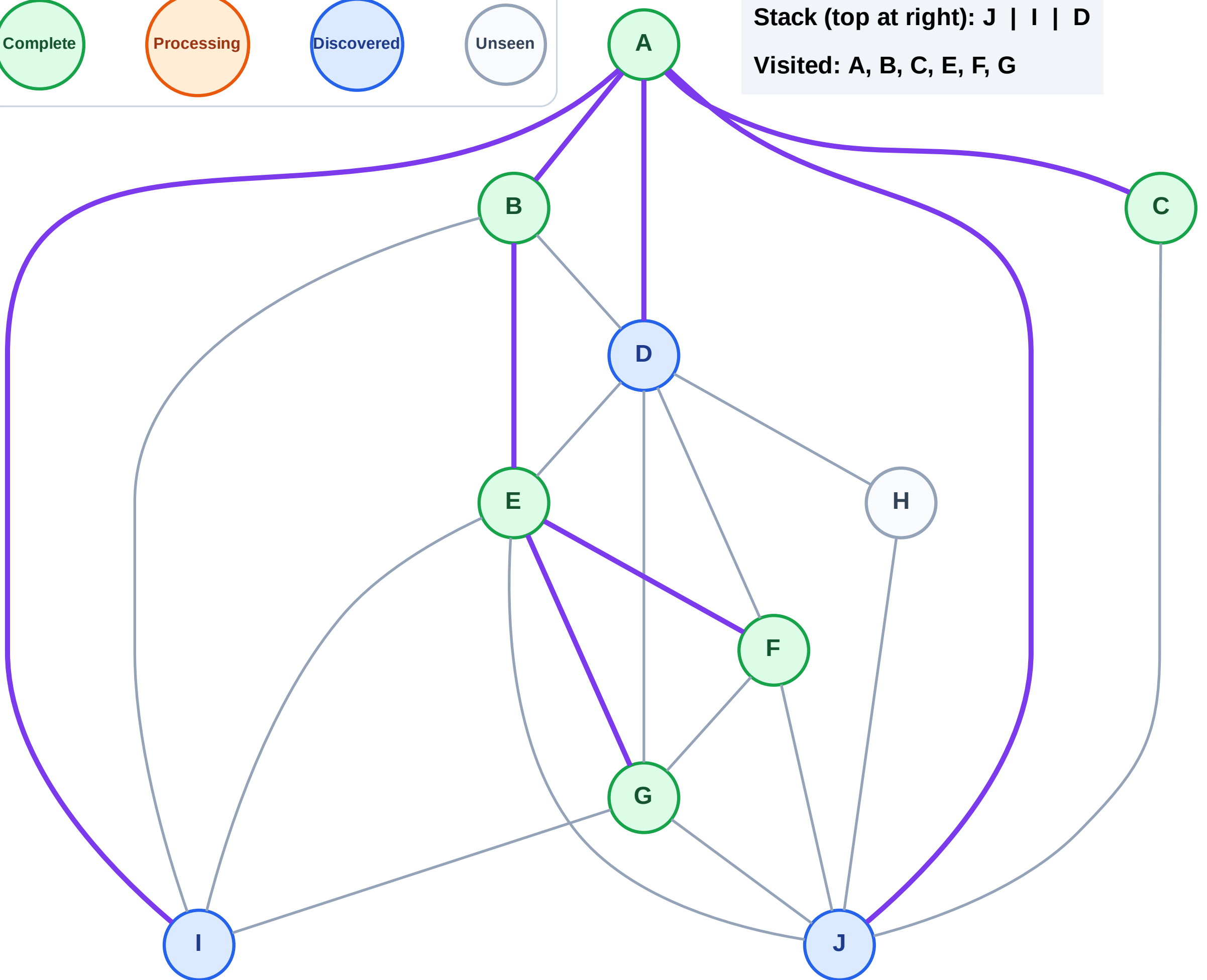
Discovered



Unseen

Stack (top at right): J | I | D

Visited: A, B, C, E, F, G



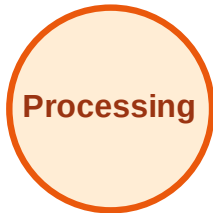
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



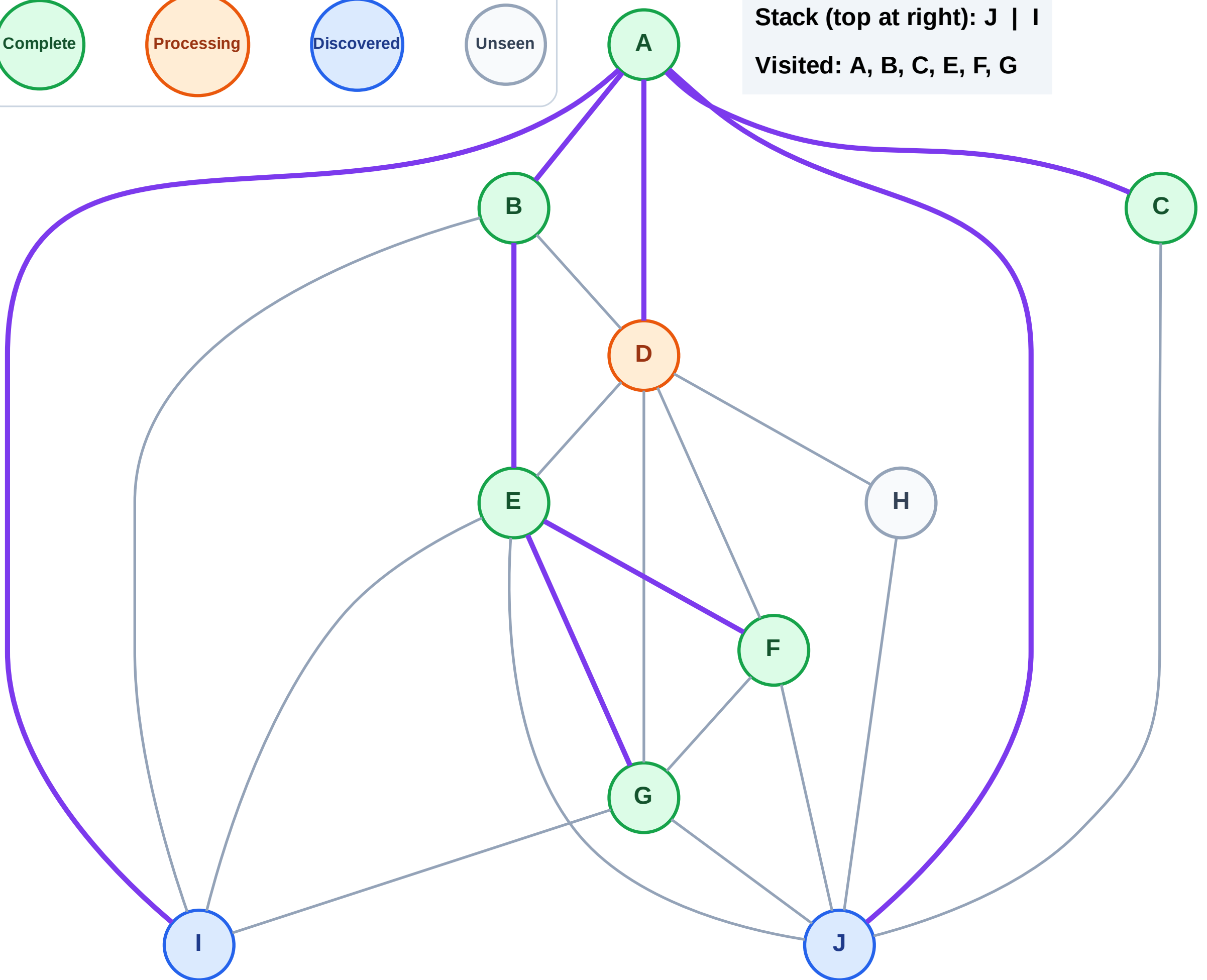
Discovered



Unseen

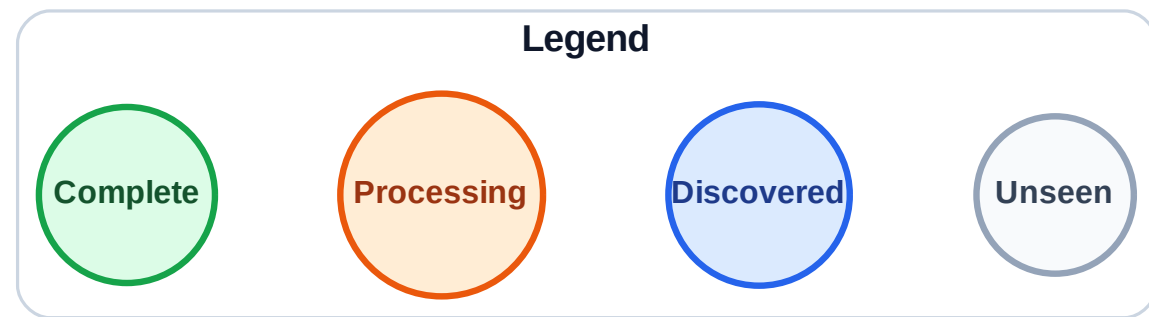
Stack (top at right): J | I

Visited: A, B, C, E, F, G

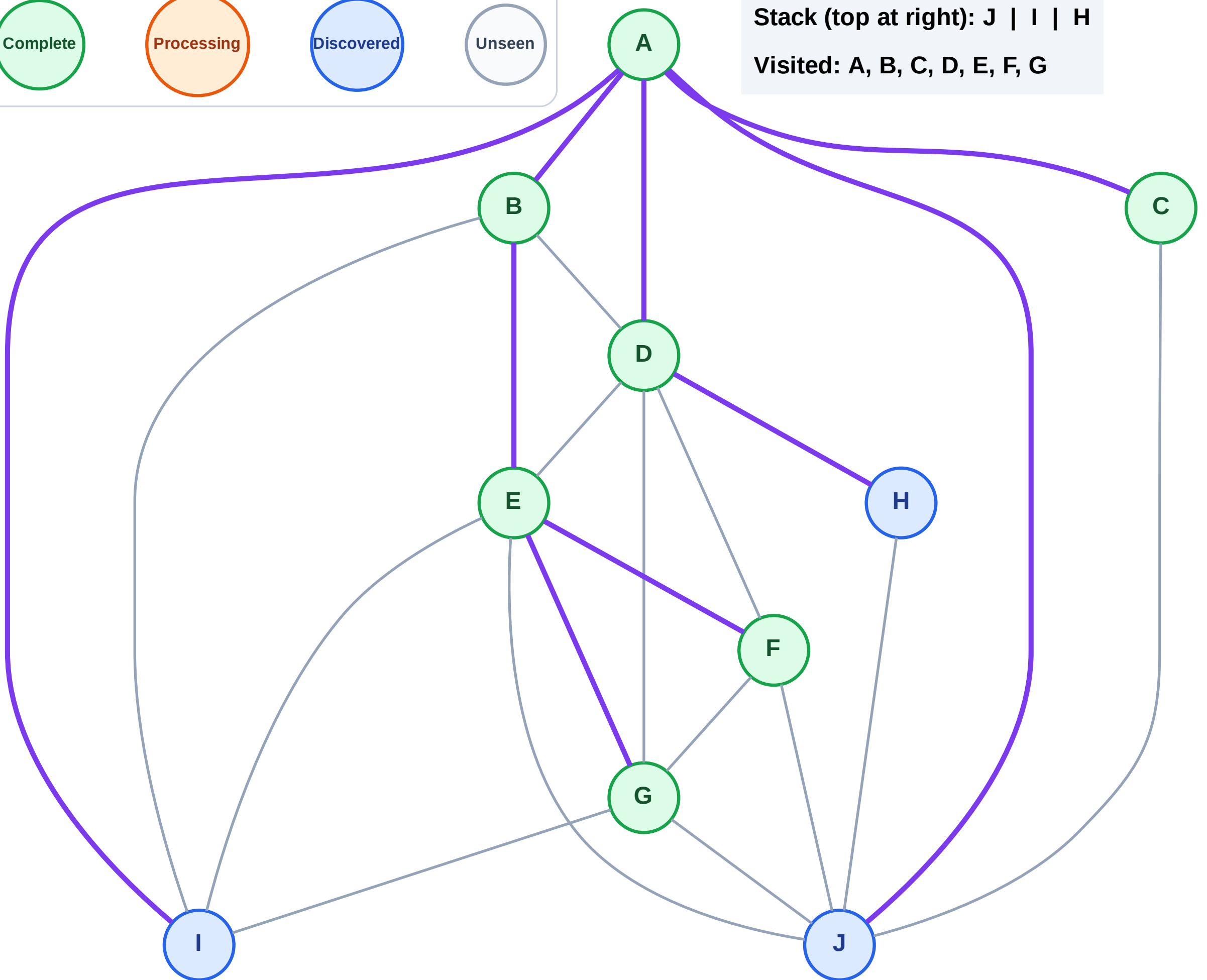


DFS Traversal

Step 15: Discover H from D



Stack (top at right): J | I | H
Visited: A, B, C, D, E, F, G



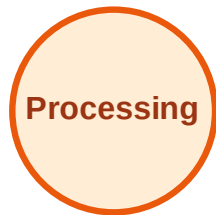
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing



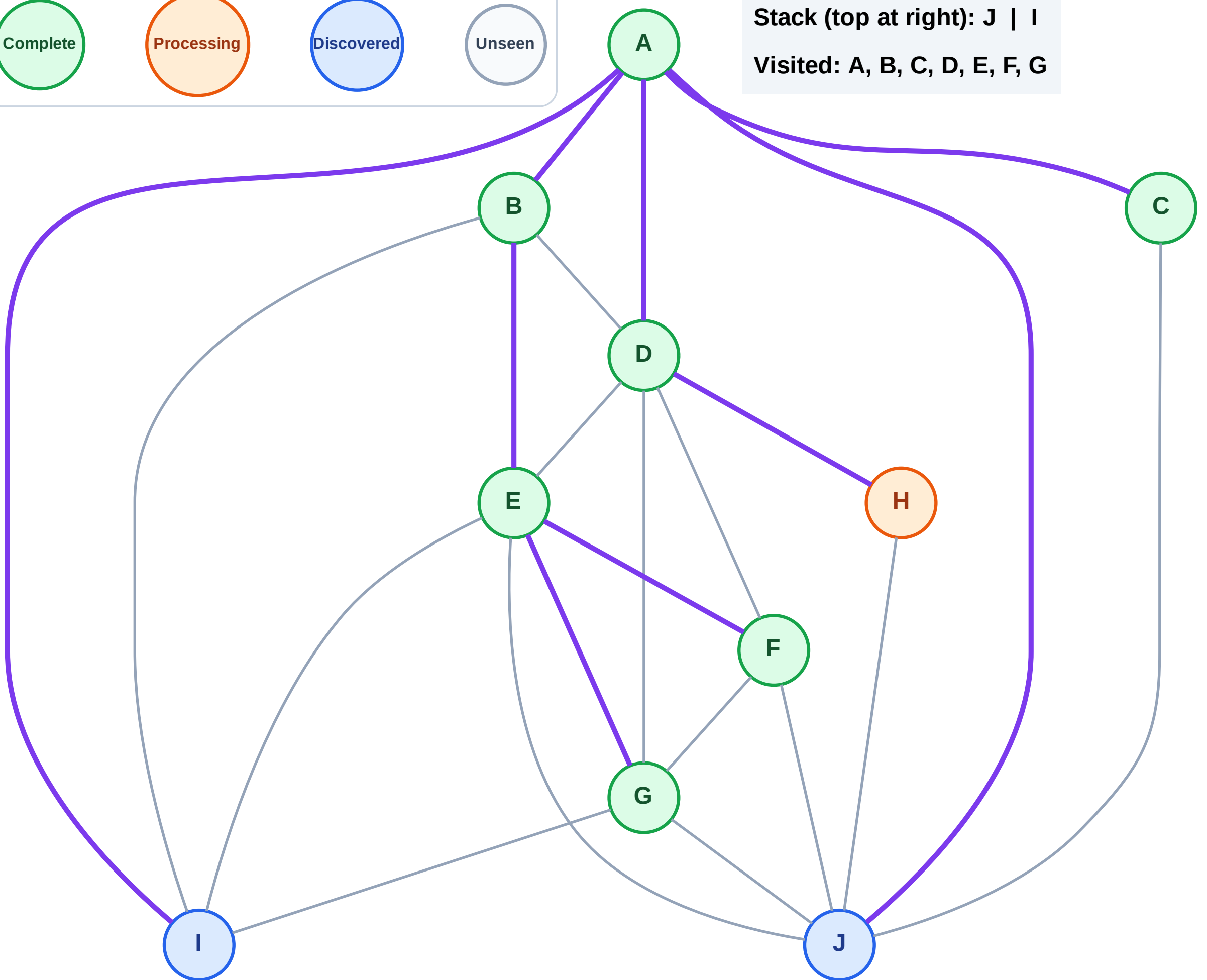
Discovered



Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G



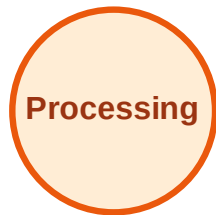
DFS Traversal

Step 17: No new neighbors from H

Legend



Complete



Processing



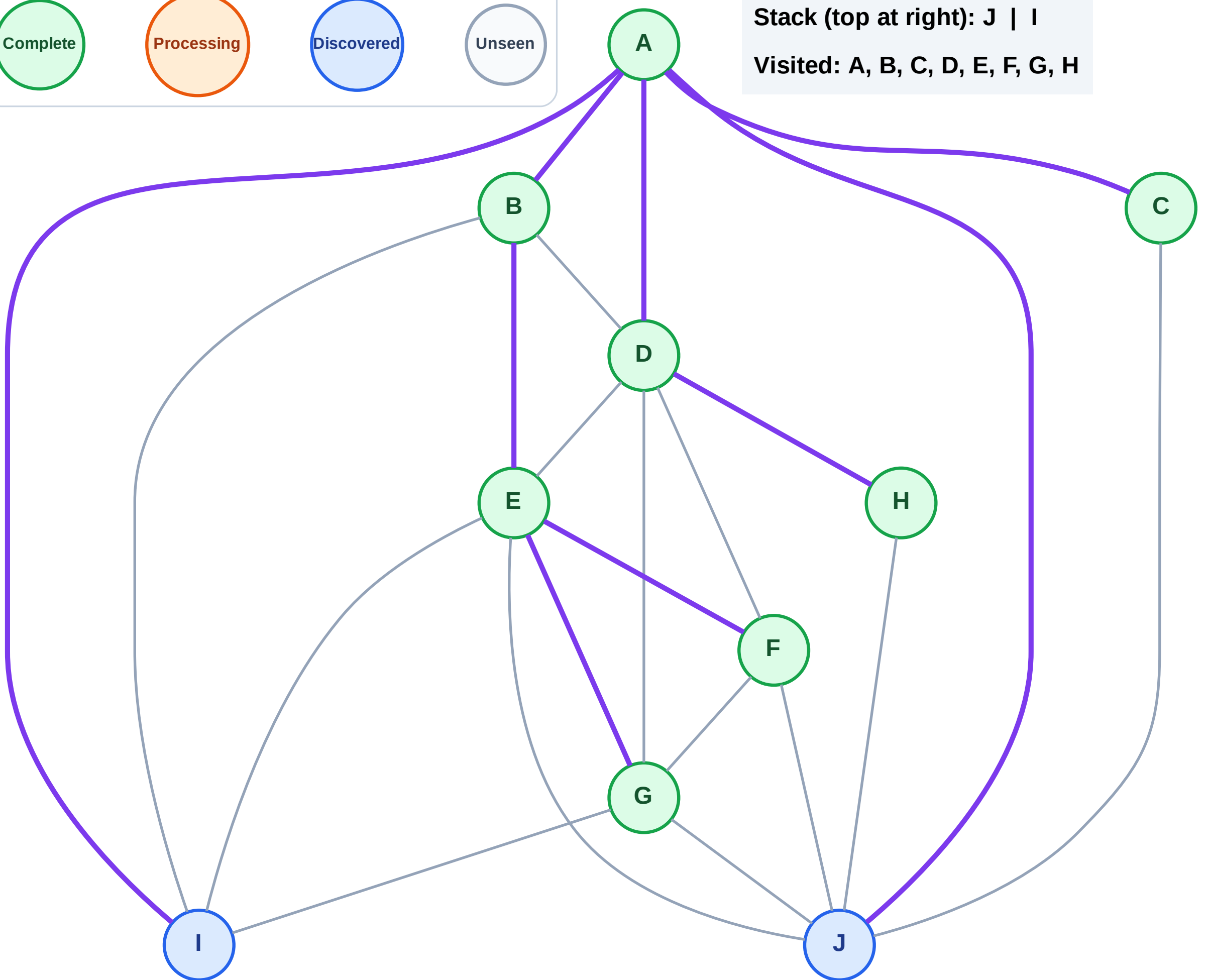
Discovered



Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



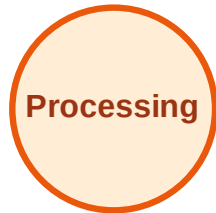
DFS Traversal

Step 18: Process node I

Legend



Complete



Processing



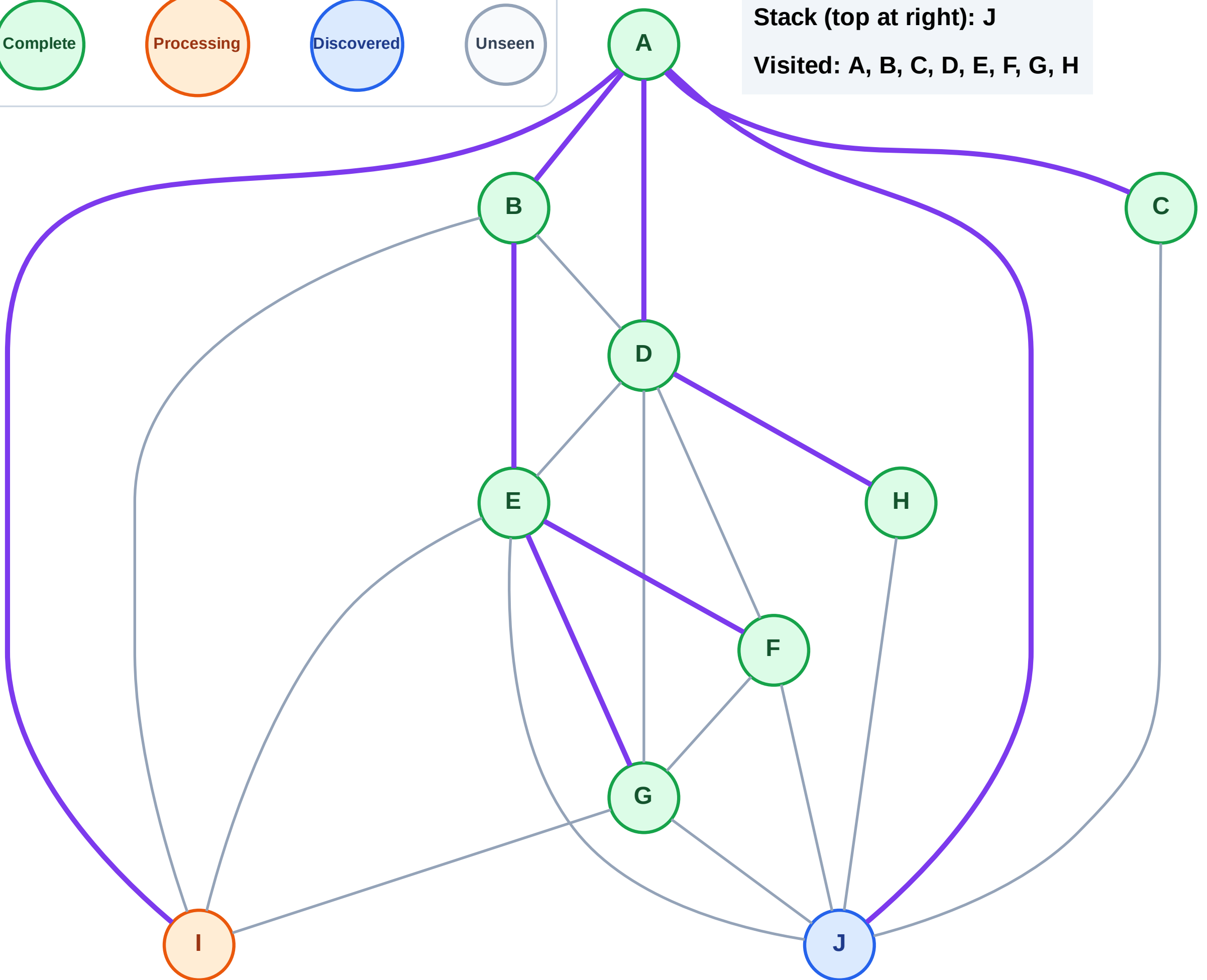
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



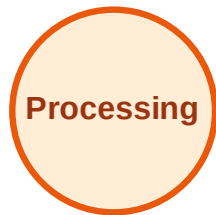
DFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



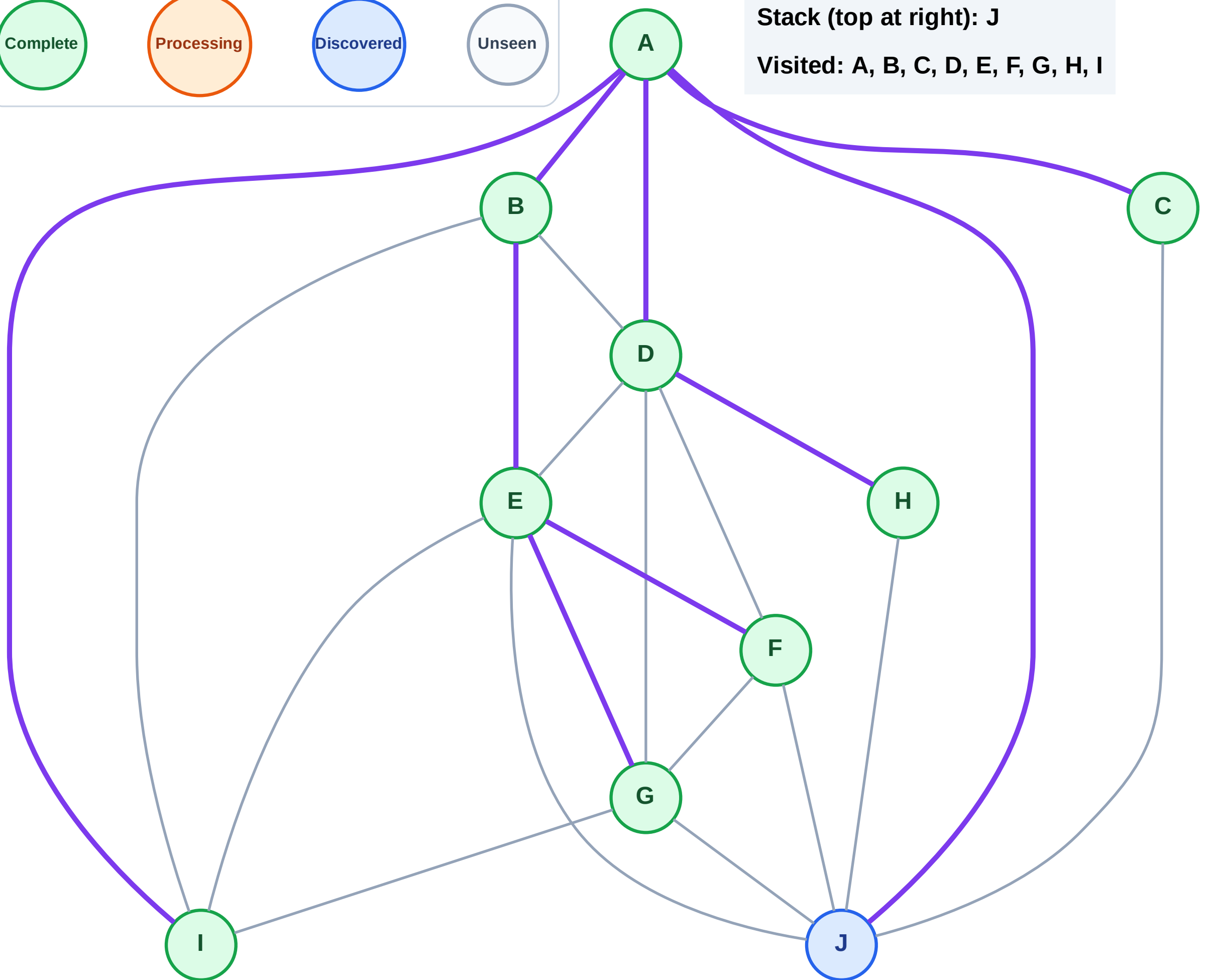
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

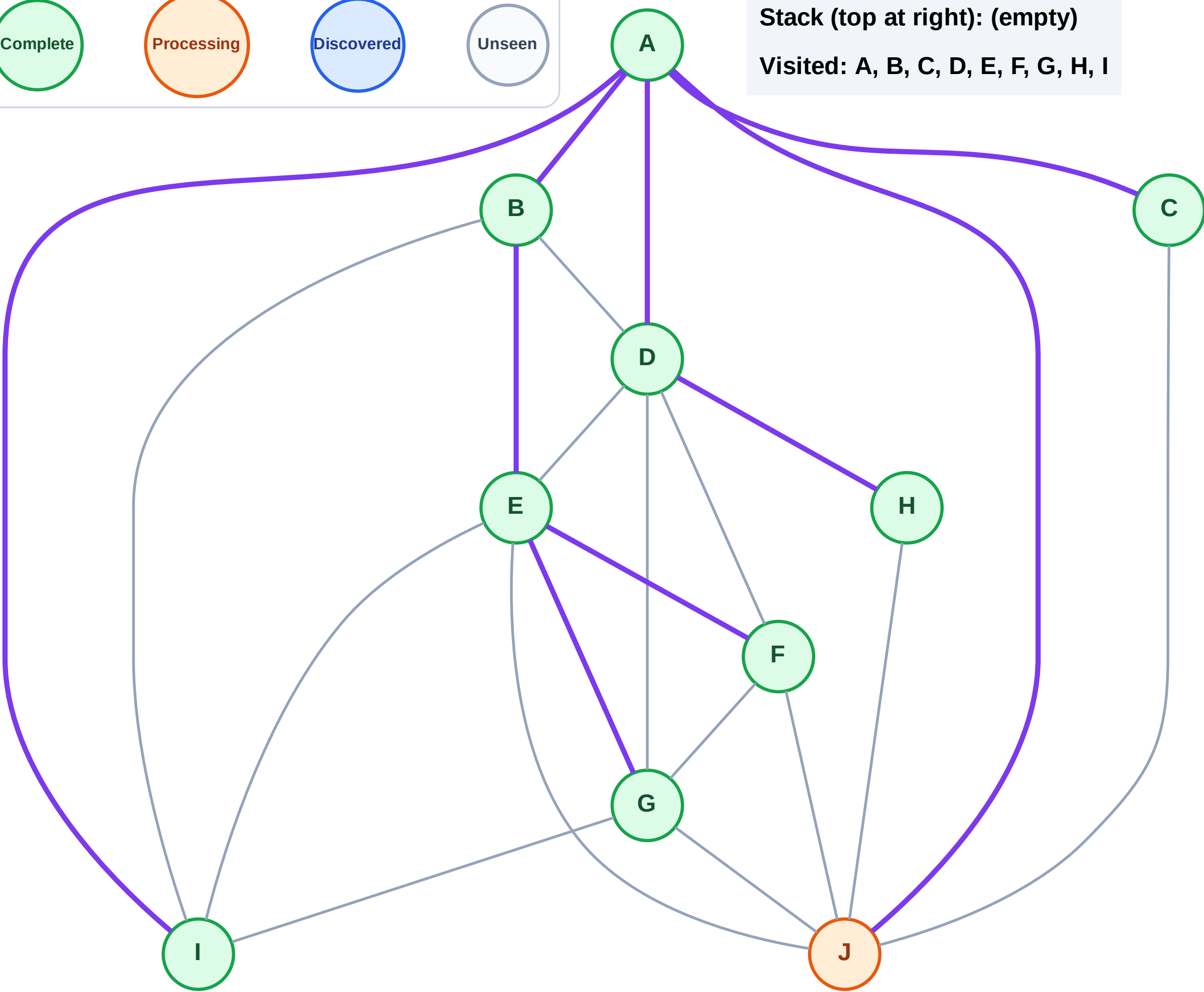
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



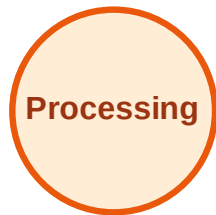
DFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

